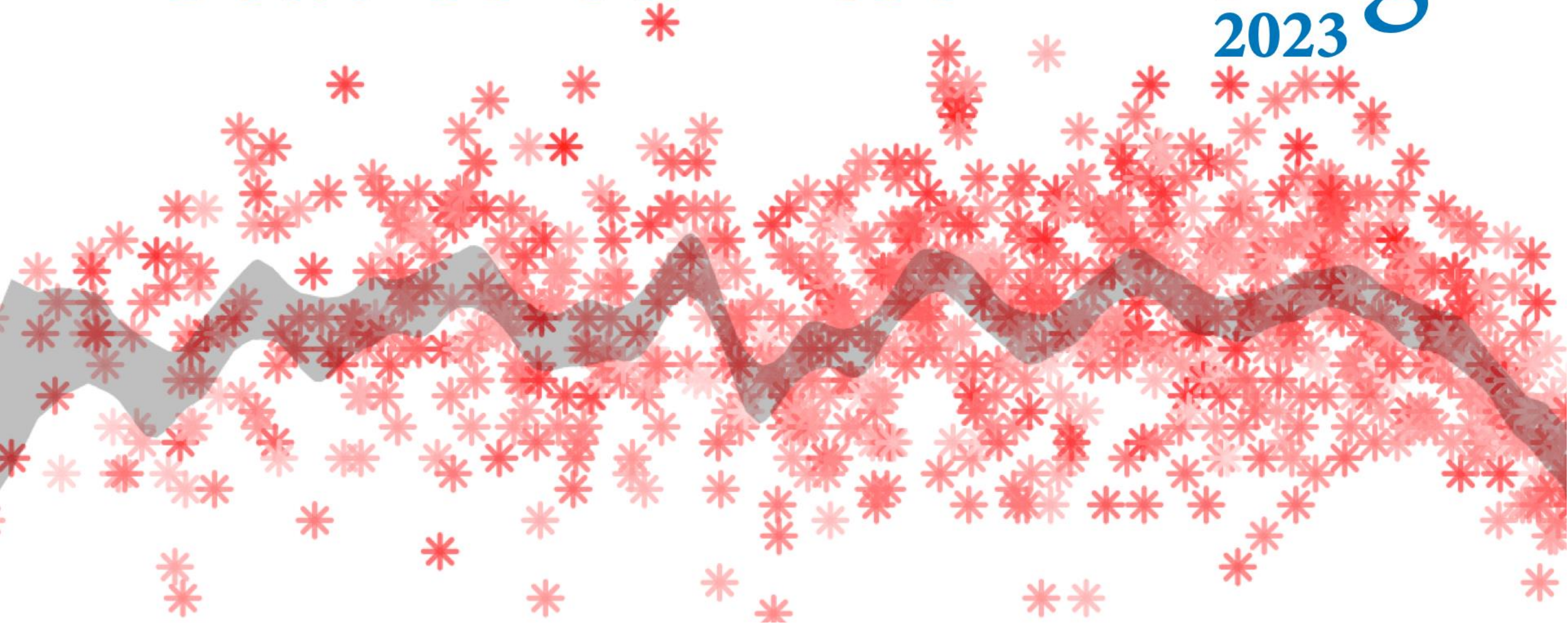


Statistical Rethinking

2023



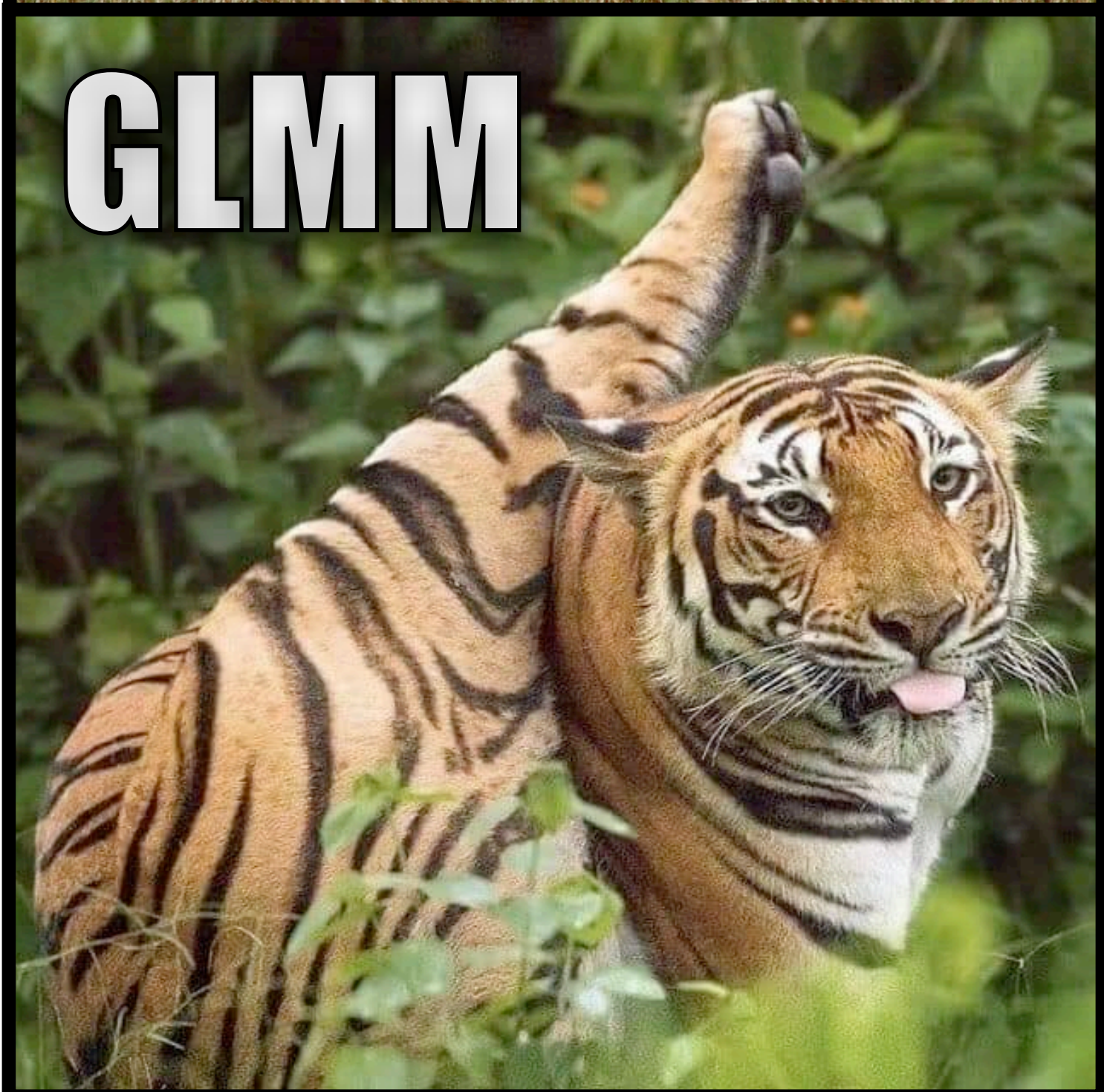
Generalized Linear Habits

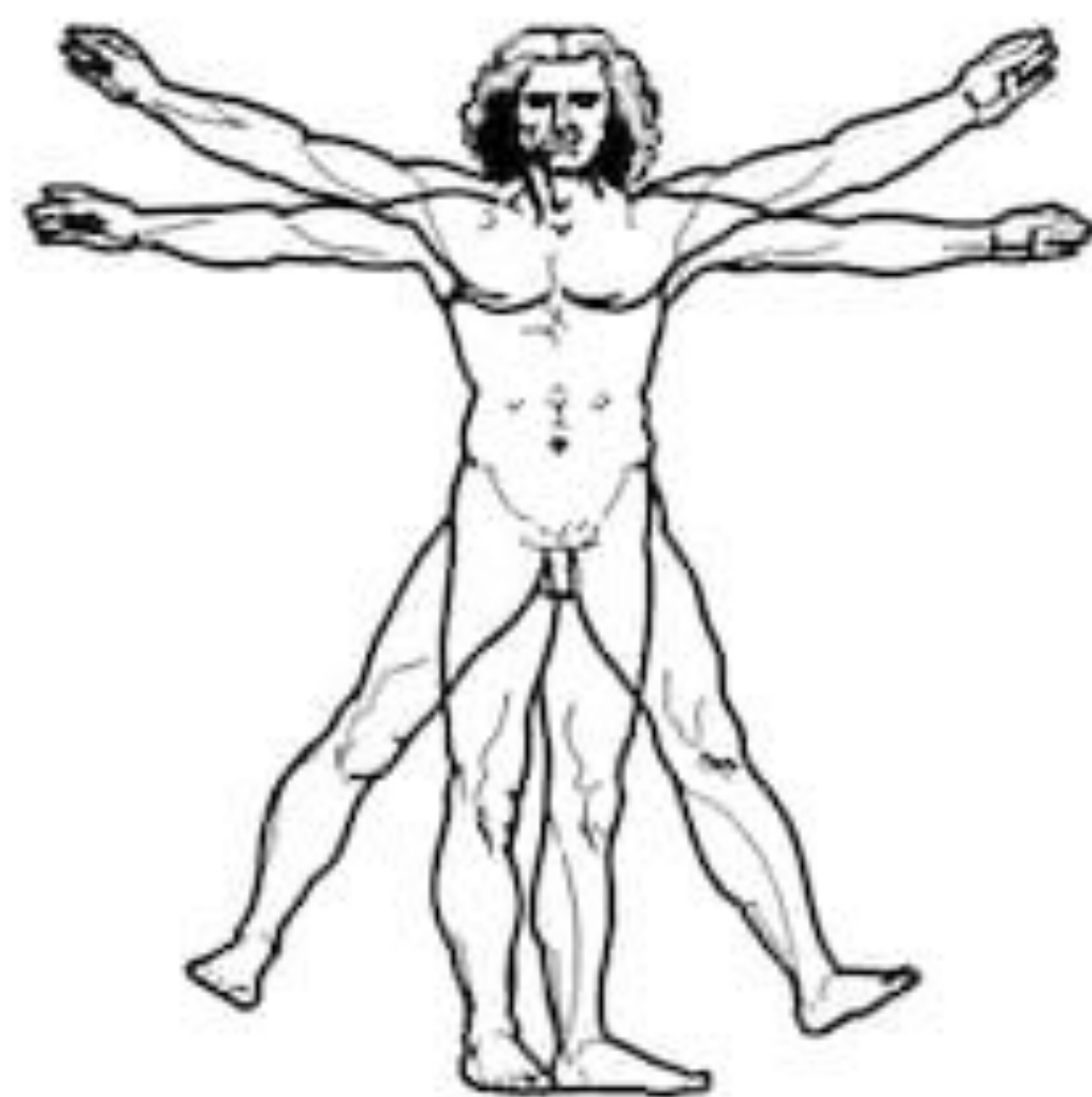
GLMs and GLMMs: Flexible association description machines

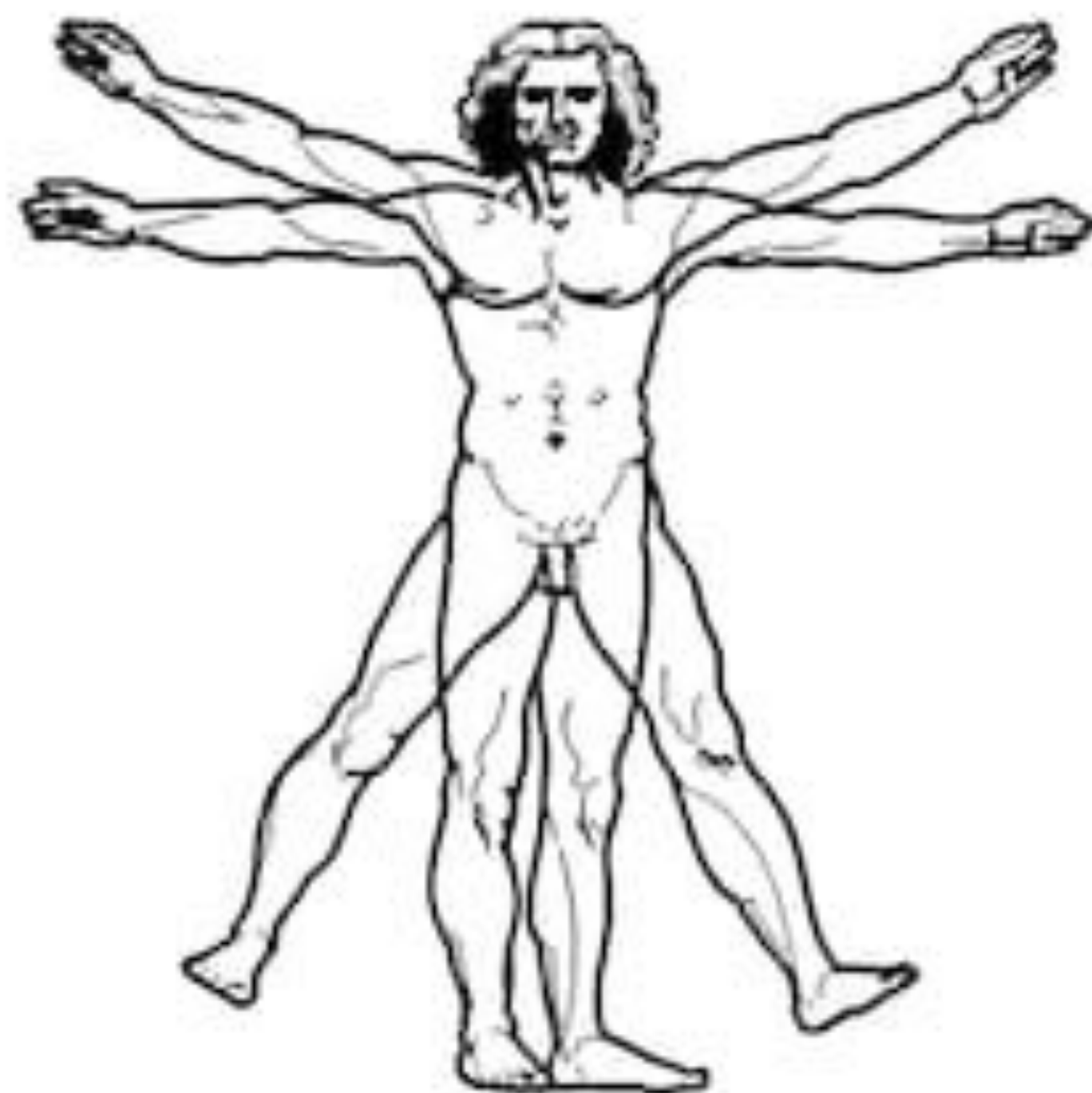
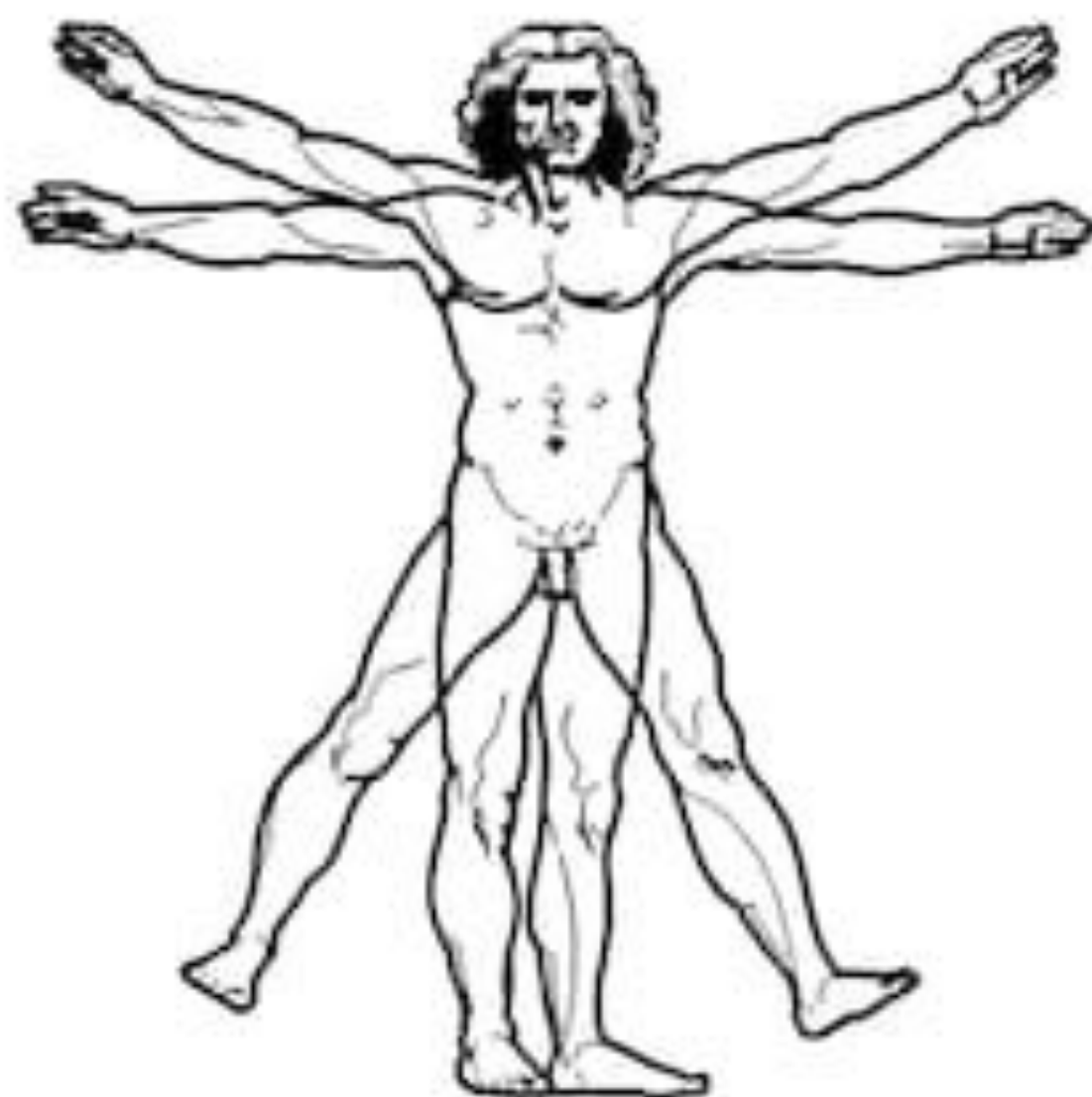
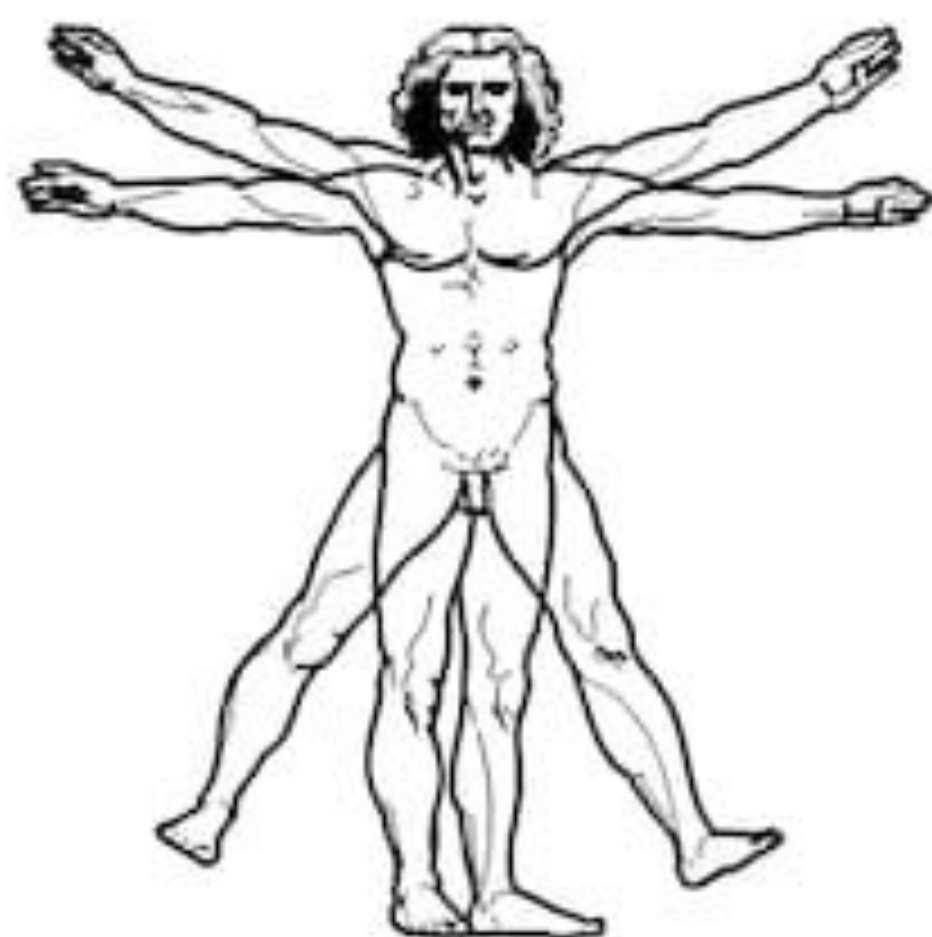
With external causal model, causal interpretation possible

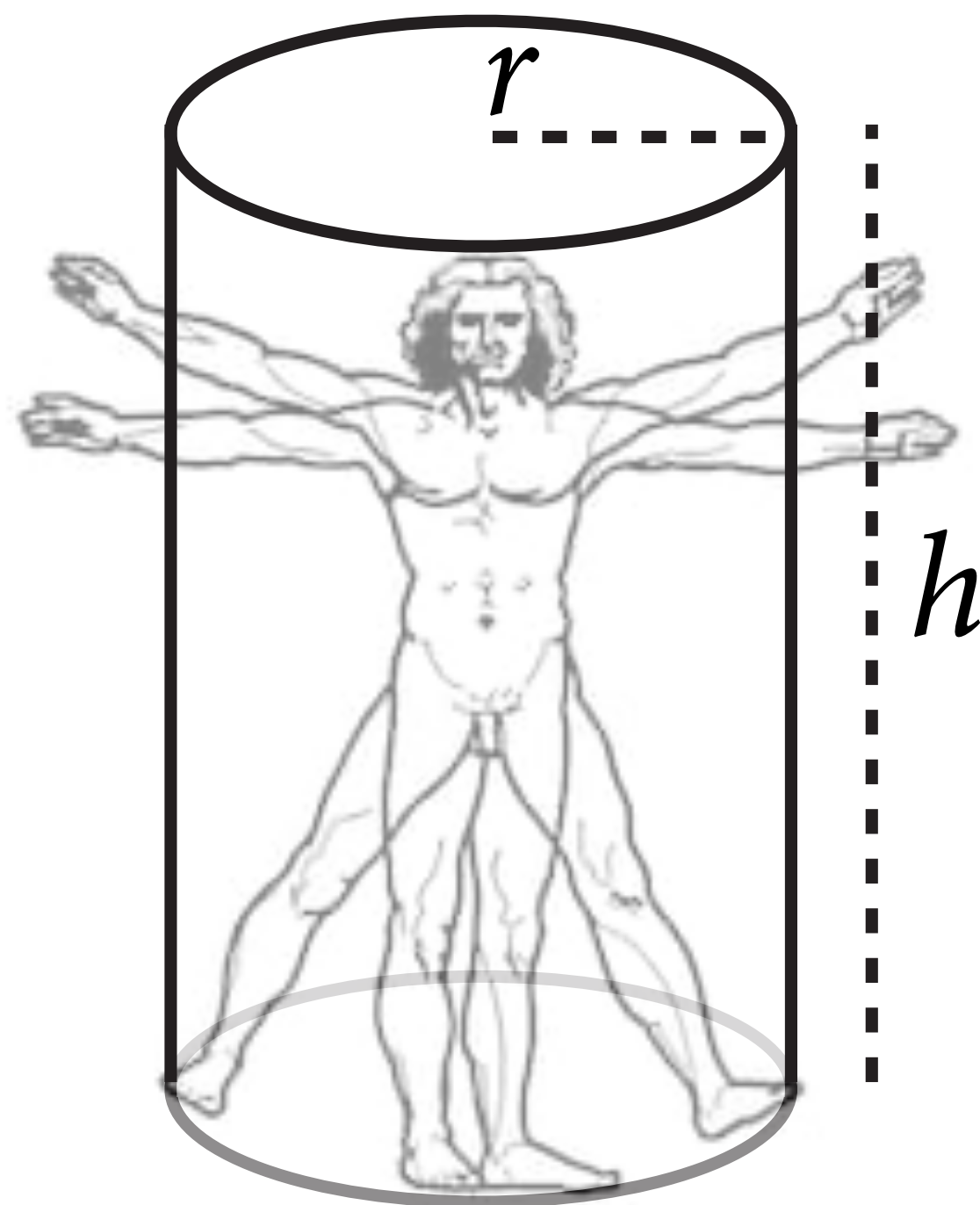
But only a fraction of scientific phenomena expressible as GLM(M)s

Even when GLM(M)s sufficient, starting with theory solves empirical problems



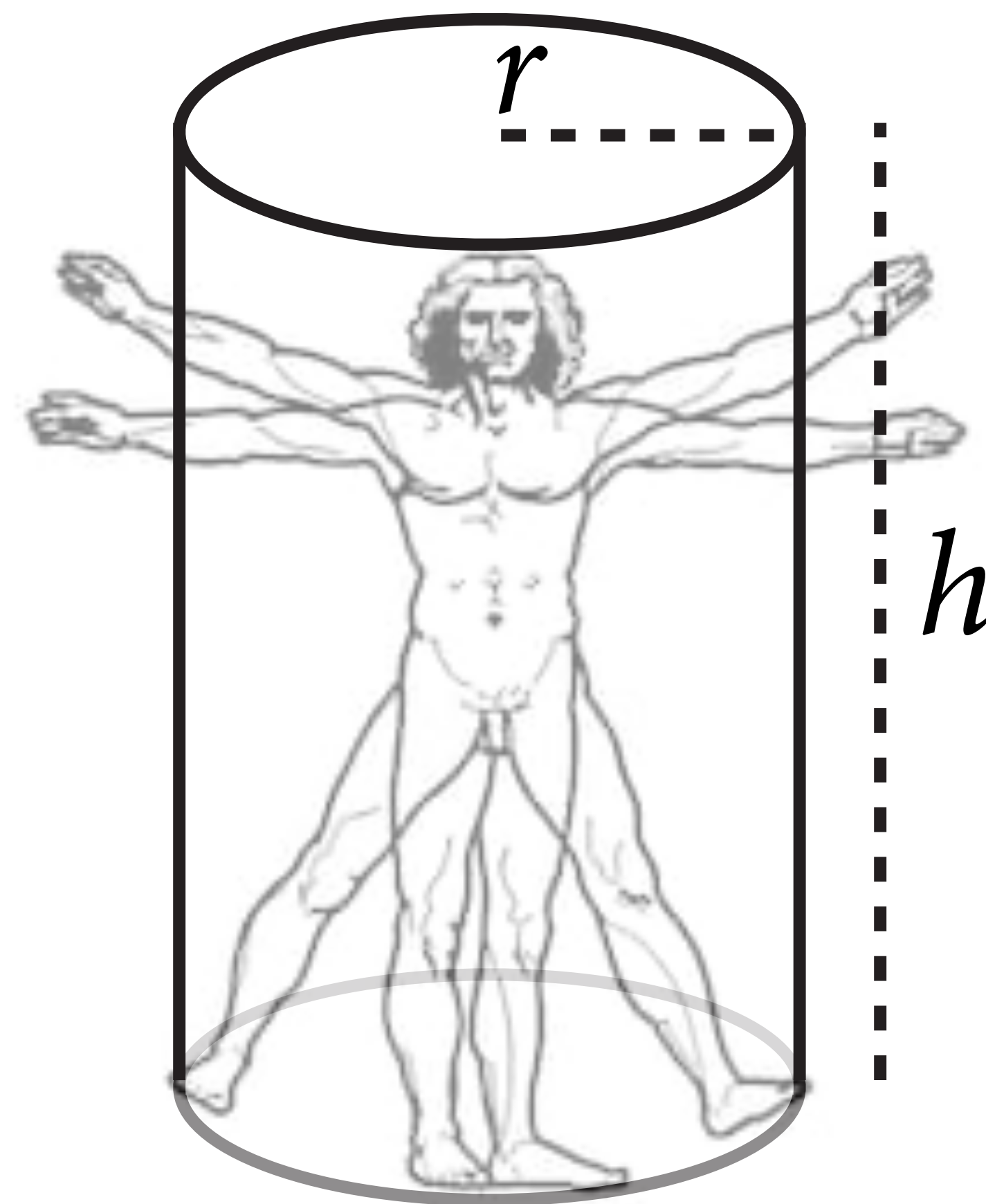






$$V = \pi r^2 h$$

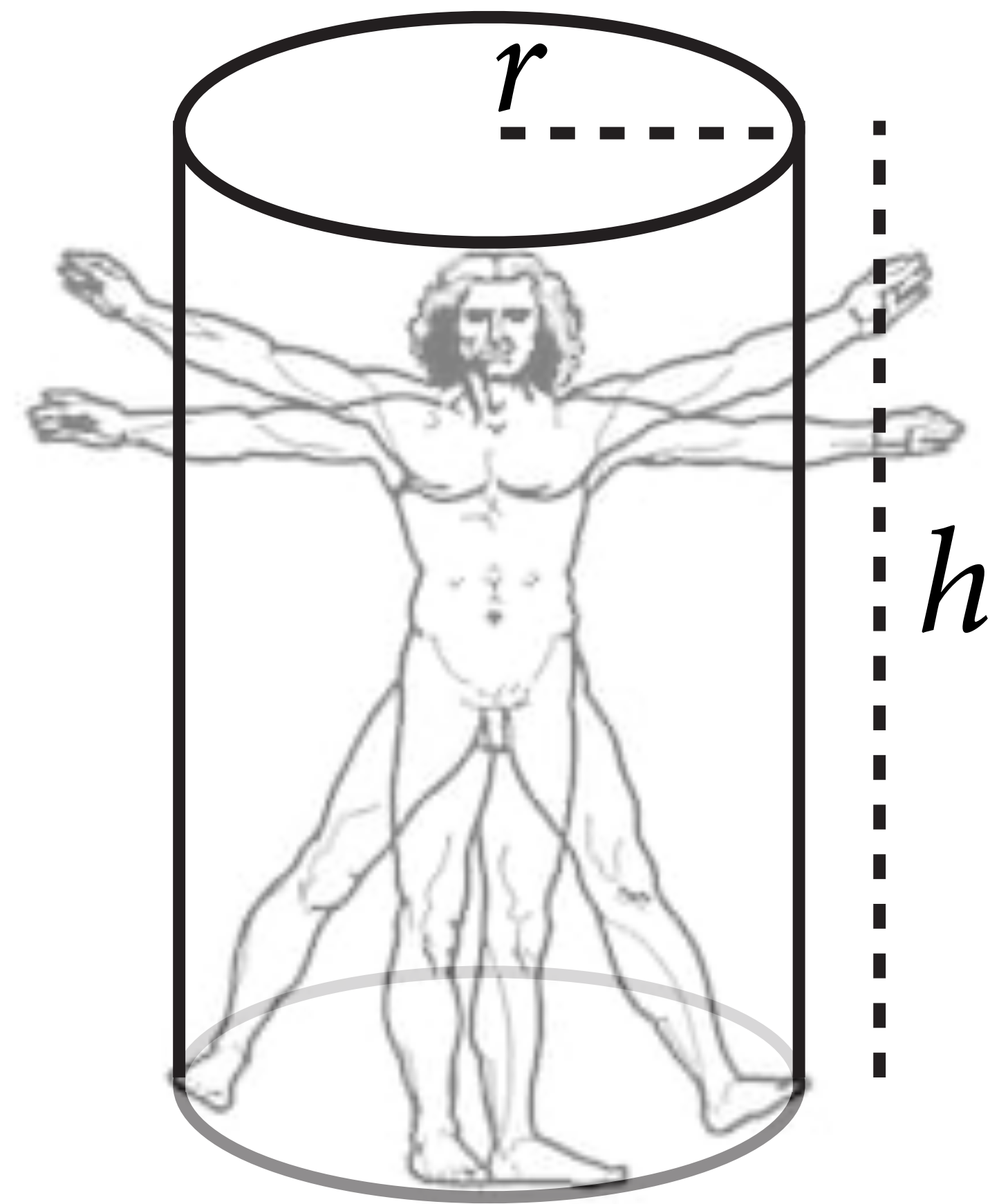
volume *radius* *height*



$$V = \pi r^2 h$$

$$V = \pi (\underline{ph})^2 h$$

*radius as
proportion of height*



$$V = \pi r^2 h$$

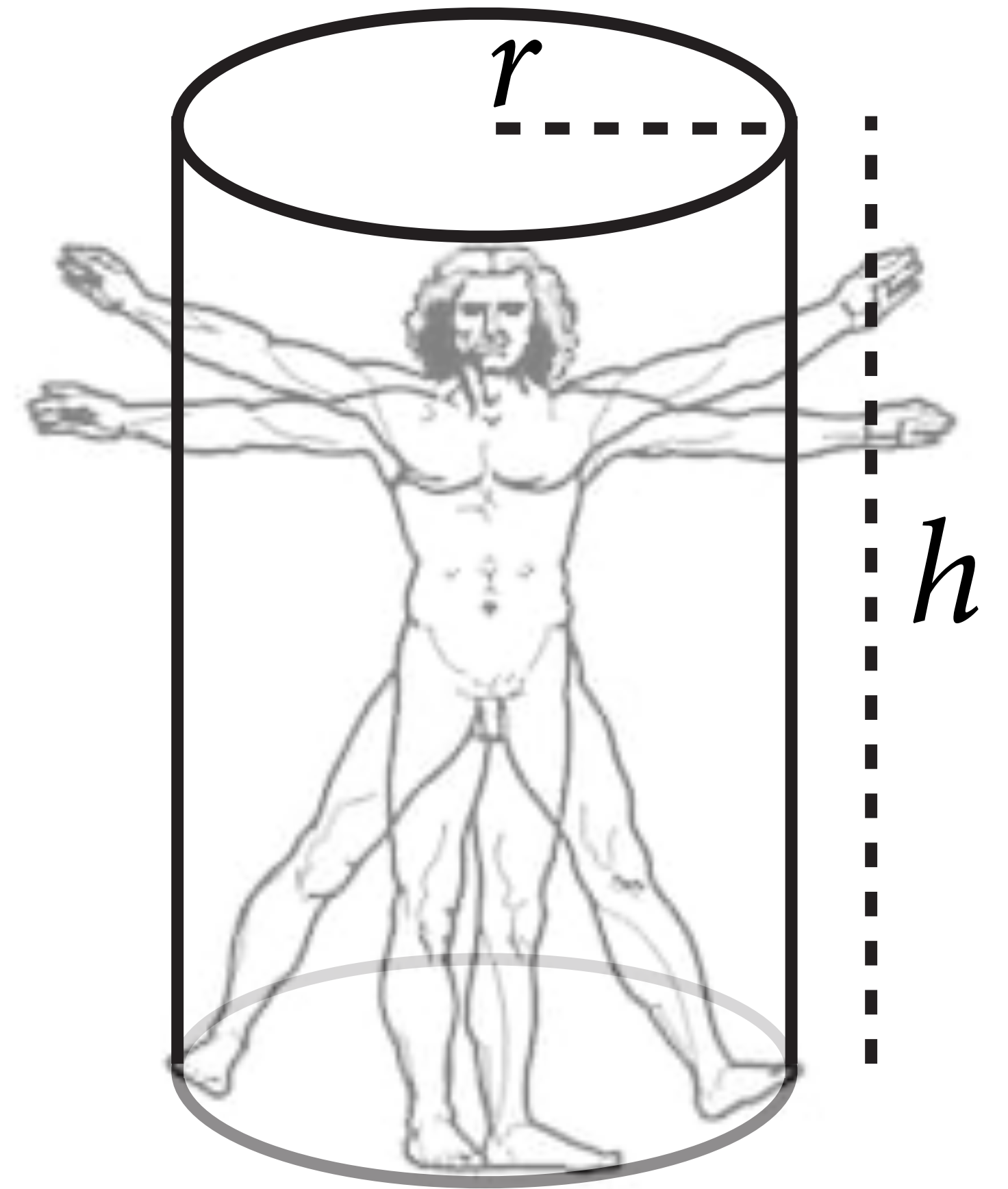
$$V = \pi (ph)^2 h$$

$$W = kV = k\pi(ph)^2 h$$

weight points to W

“density” points to the first k

“density” points to the second k

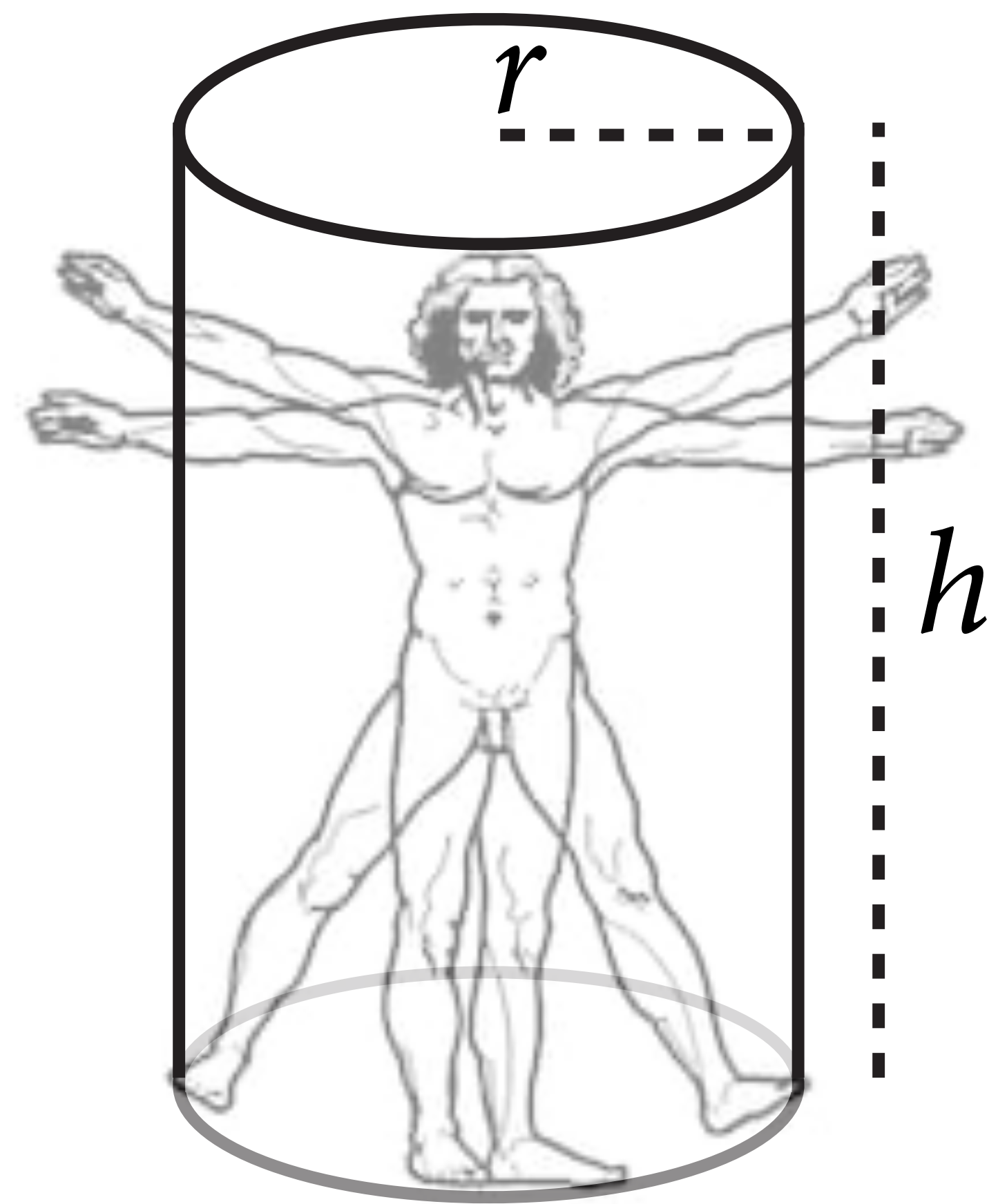


$$V = \pi r^2 h$$

$$V = \pi (ph)^2 h$$

$$W = kV = k\pi (ph)^2 h$$

$$W = k\pi p^2 h^3$$



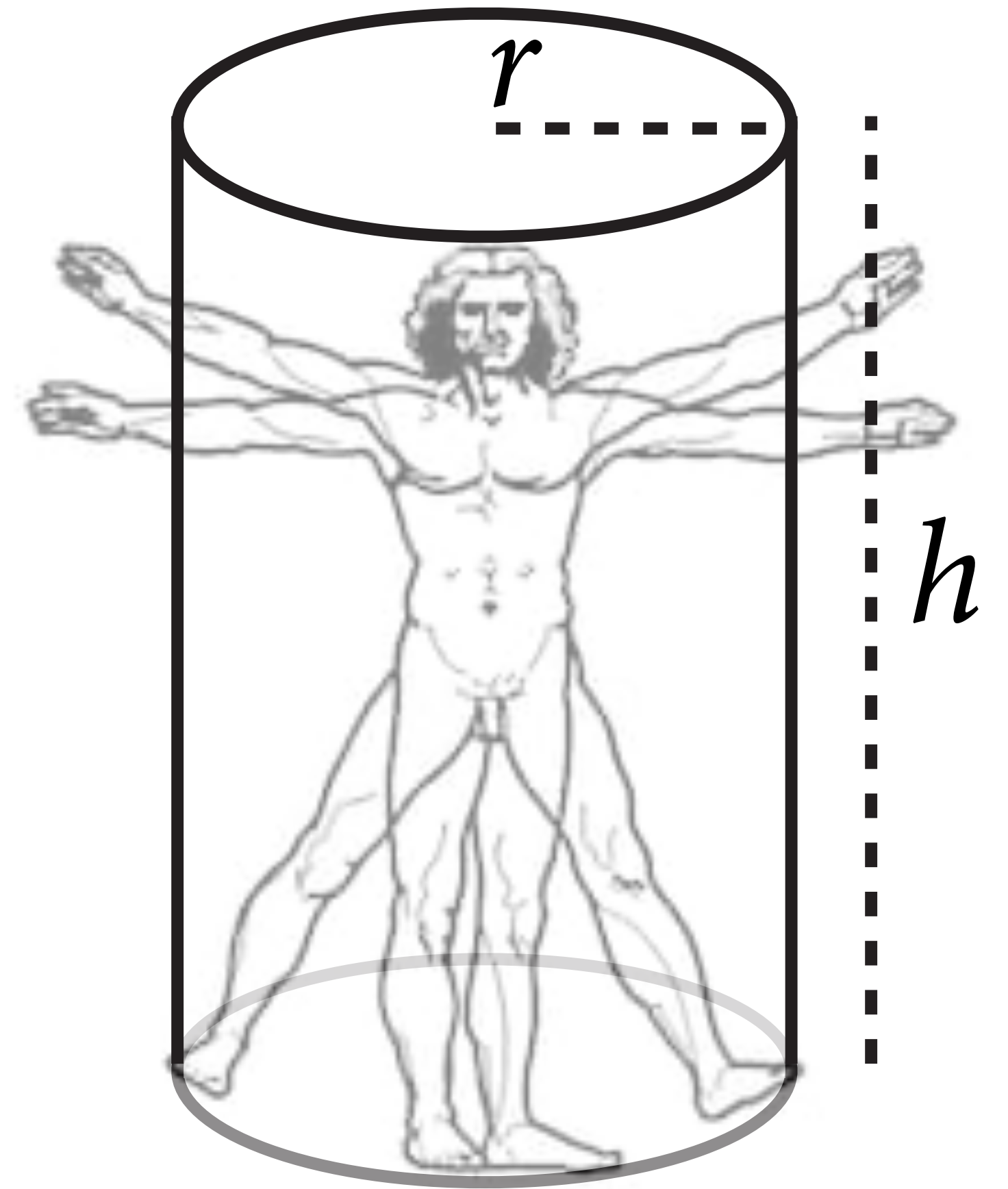
weight (data)

height (data)

$$W = k\pi r^2 h^3$$

density

proportionality



$$W_i \sim \text{Distribution}(\mu_i, \dots)$$

“error” distribution for W

$$\mu_i = k\pi p^2 H_i^3$$

expected W for H

$$p \sim \text{Distribution}(\dots)$$

prior for proportionality

$$k \sim \text{Distribution}(\dots)$$

prior for density

How to set these priors?

(1) Choose measurement scales

(2) Simulate

(3) Think

$p \sim \text{Distribution}(\dots)$

prior for proportionality

$k \sim \text{Distribution}(\dots)$

prior for density

How to set these priors?

(1) Choose measurement scales

(2) Simulate

(3) Think

$\mu_i = k \pi \rho^2 H_i^3$

kg *kg/cm³* *cm³*

unitless ratios

How to set these priors?

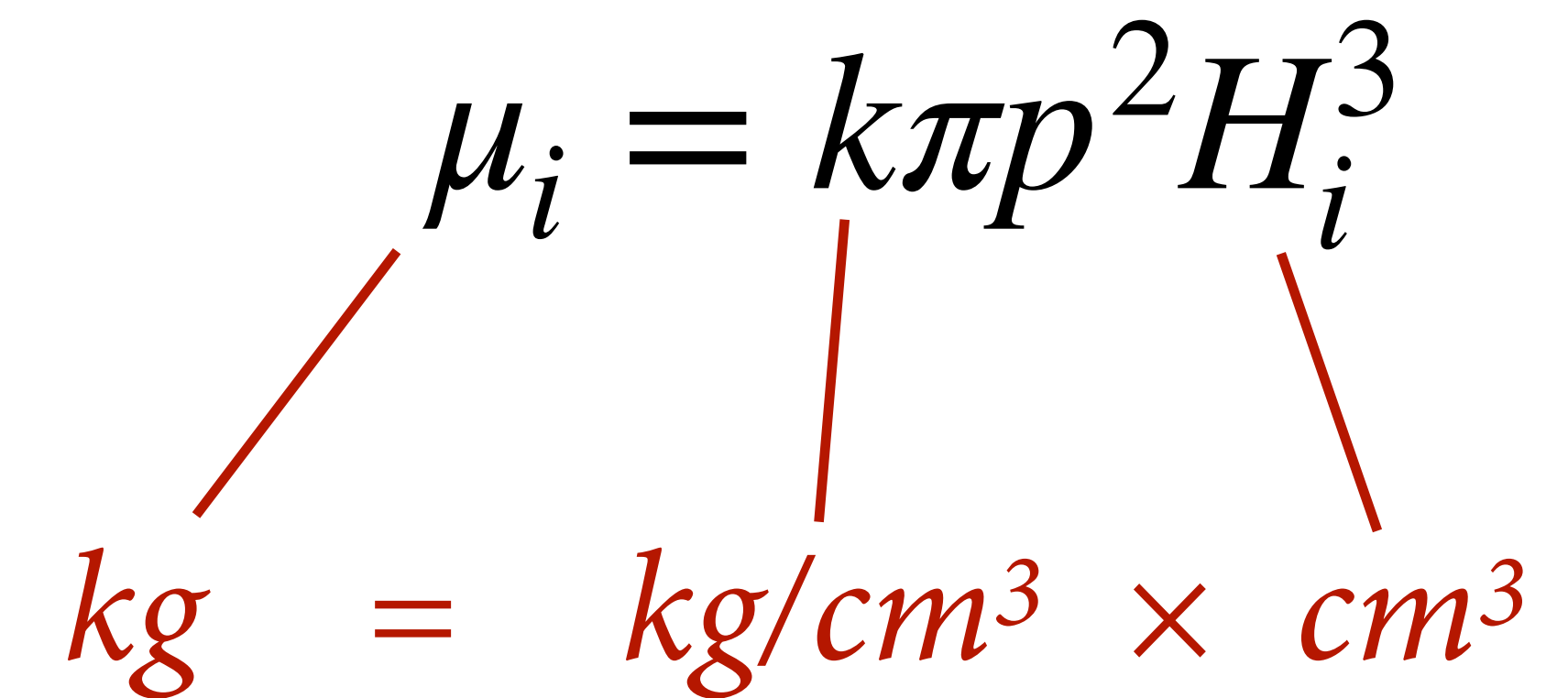
(1) Choose measurement scales

(2) Simulate

(3) Think

$$\mu_i = k\pi r^2 H_i^3$$

$kg = kg/cm^3 \times cm^3$

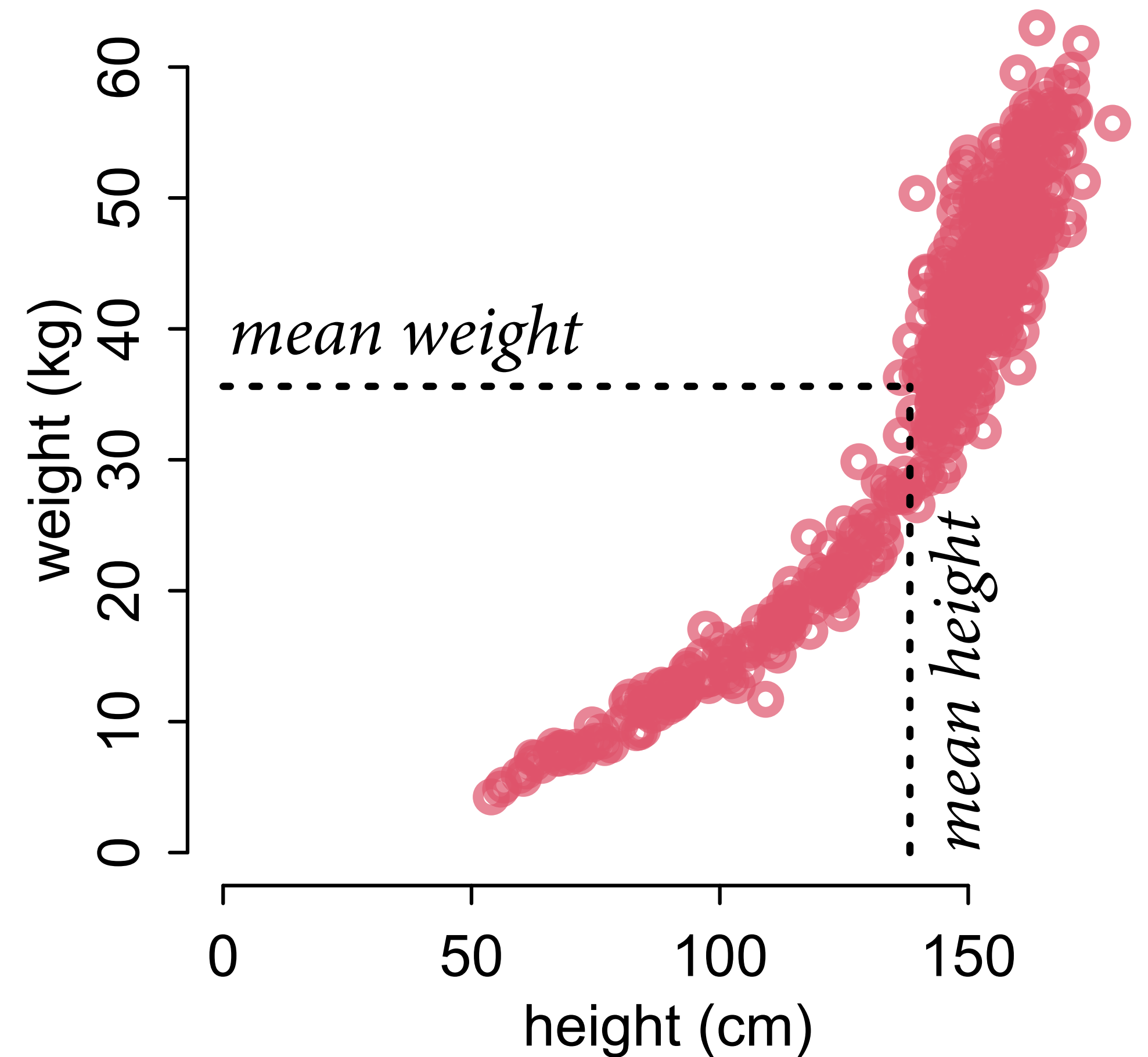


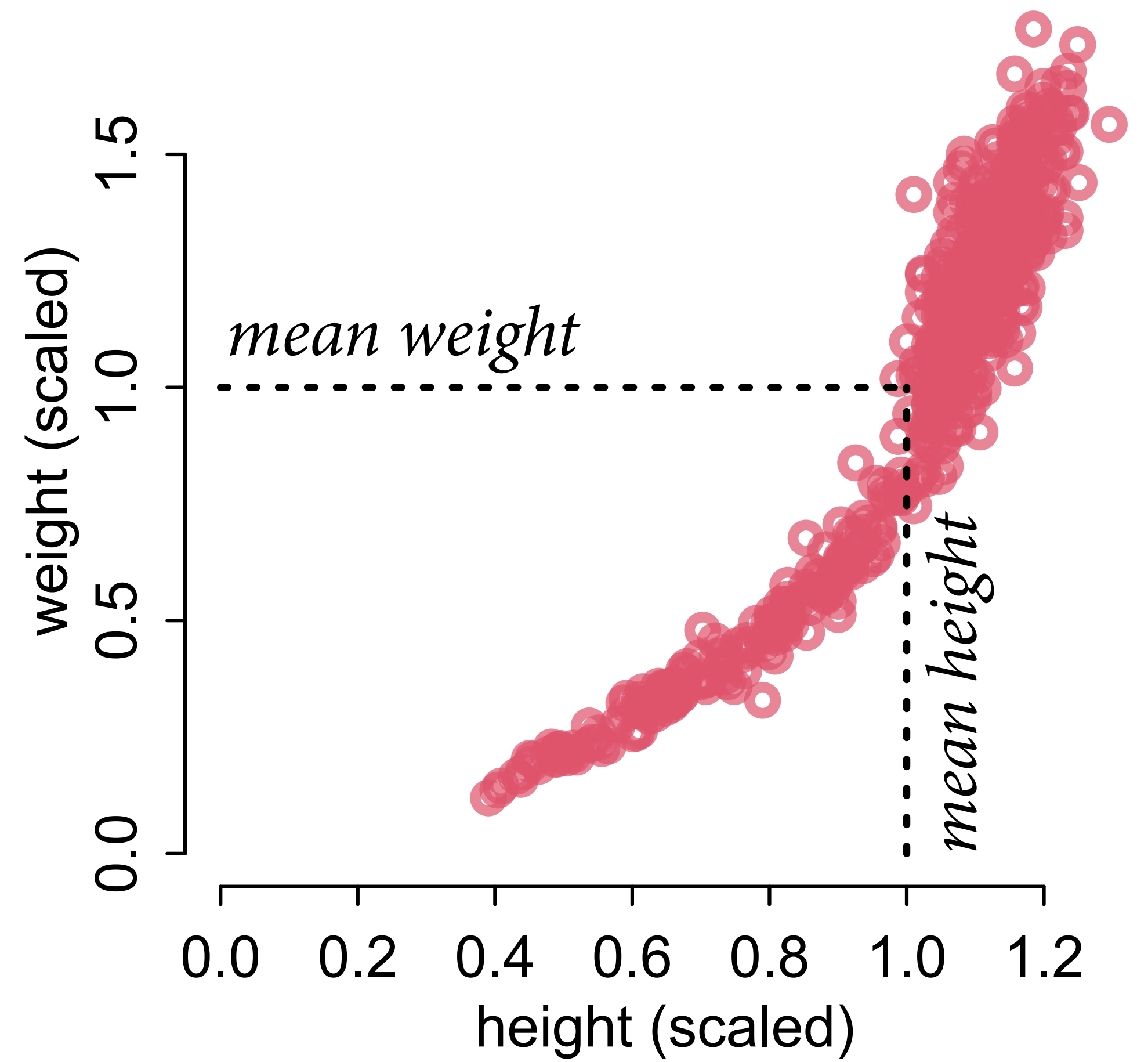
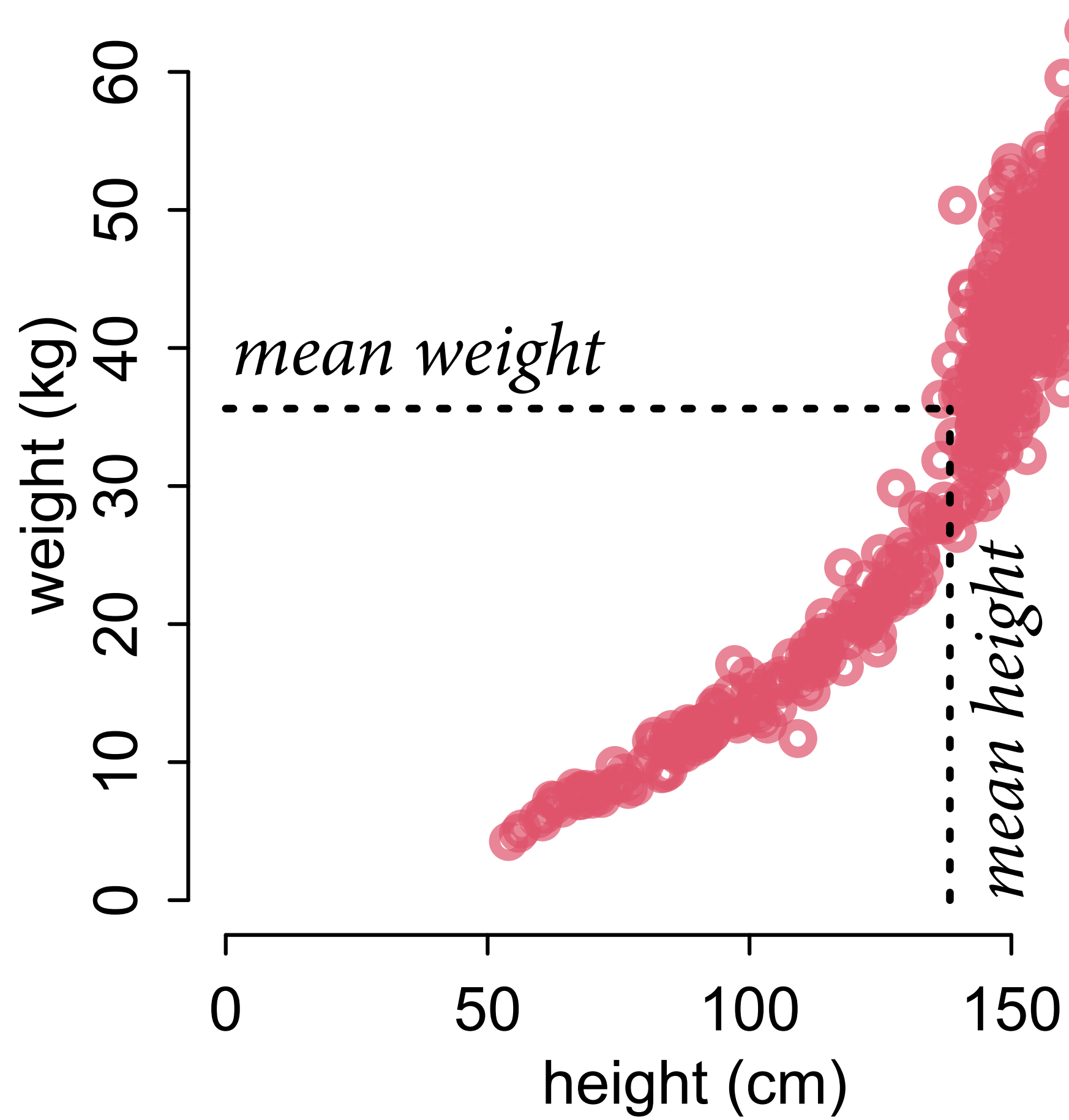
$$\mu_i = k\pi r^2 H_i^3$$

$$kg = kg/cm^3 \times cm^3$$

Measurement scales are artifice

If you can divide out all
measurement units (kg, cm),
often easier





Here, the scaling is $\text{variable} / \text{mean}(\text{variable})$

How to set these priors?

(1) Choose measurement scales

(2) *Simulate*

(3) Think

$p \sim \text{Distribution}(\dots)$

between 0–1, < 0.5

$k \sim \text{Distribution}(\dots)$

positive real, > 1

How to set these priors?

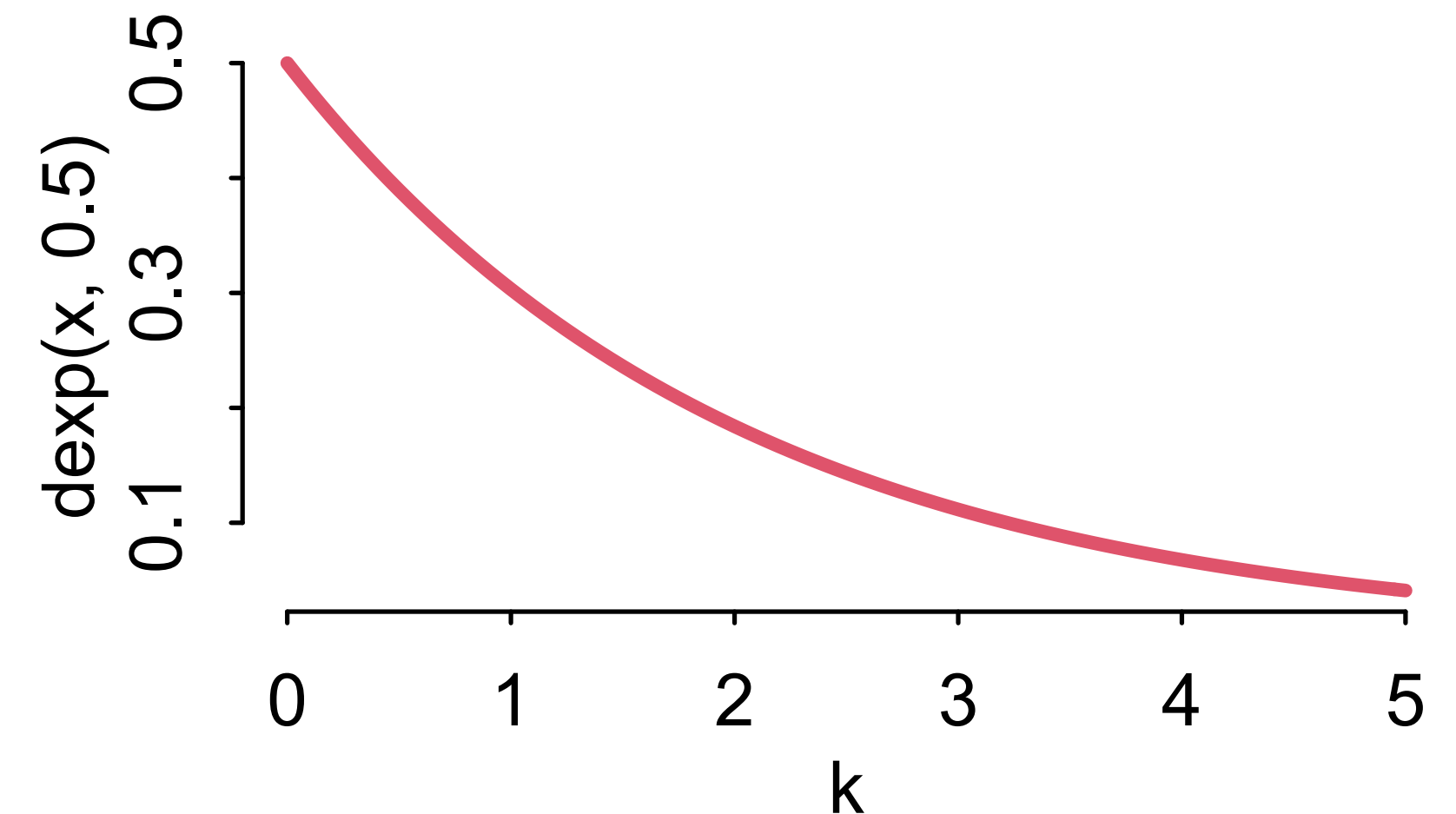
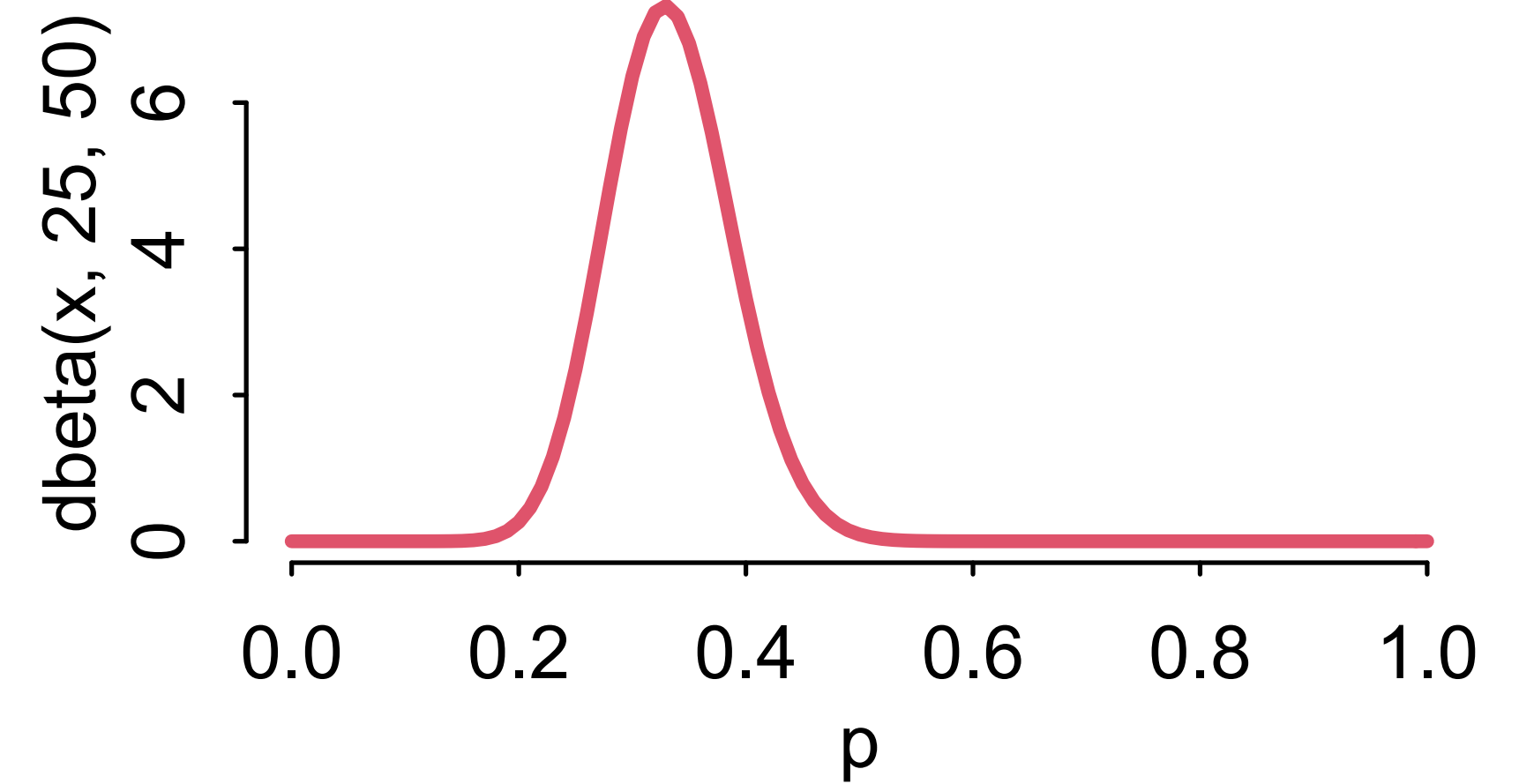
(1) Choose measurement scales

(2) Simulate

(3) Think

$$p \sim \text{Beta}(25, 50)$$

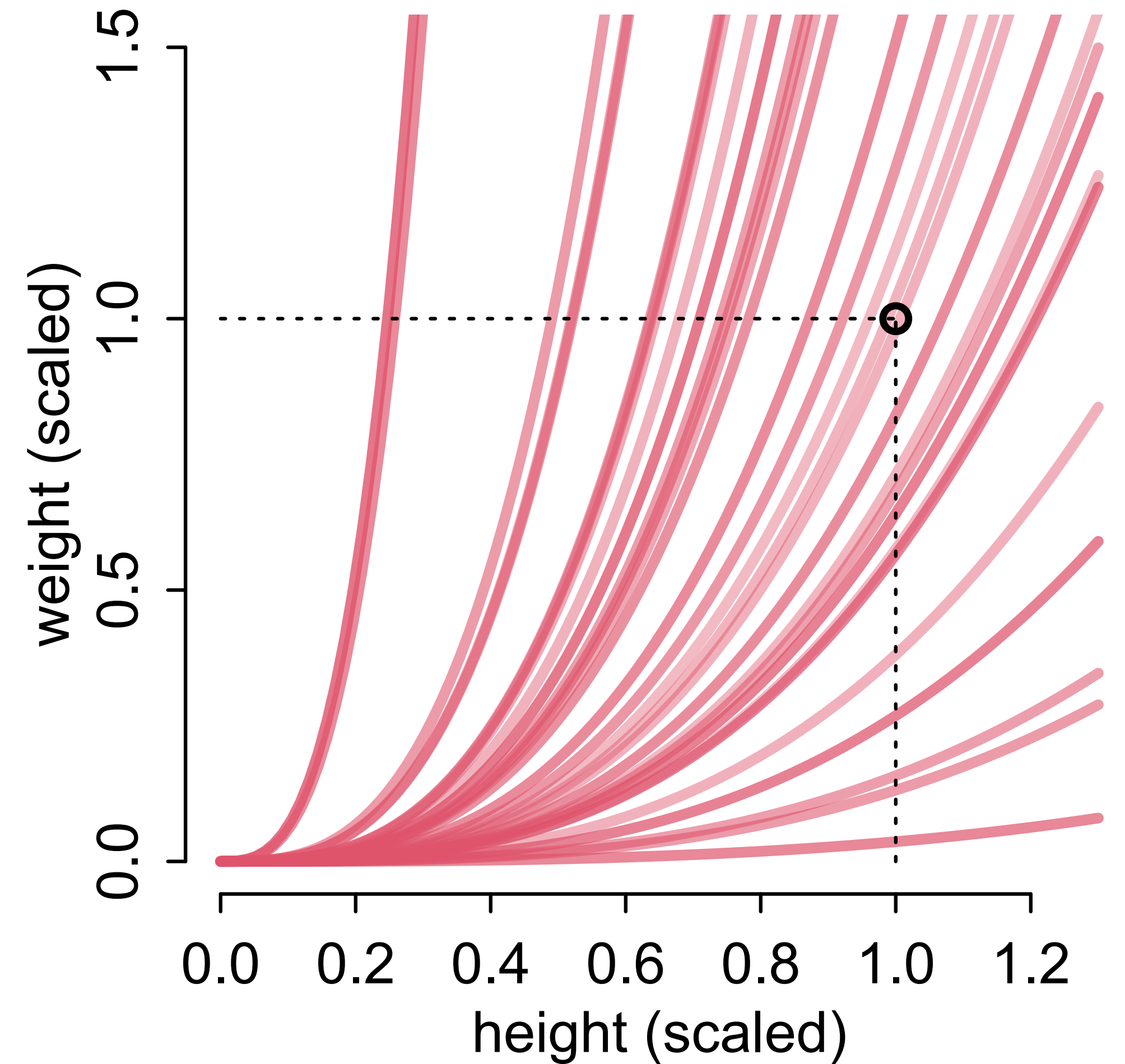
$$k \sim \text{Exponential}(0.5)$$



Prior predictive simulation

```
# prior sim
n <- 30
p <- rbeta(n,25,50)
k <- rexp(n,0.5)
sigma <- rexp(n,1)

xseq <- seq(from=0,to=1.3,len=100)
plot(NULL,xlim=c(0,1.3),ylim=c(0,1.5))
for ( i in 1:n ) {
  mu <- log( pi * k[i] * p[i]^2 * xseq^3 )
  lines( xseq , exp(mu + sigma[i]^2/2) ,
  lwd=3 , col=col.alpha(2,runif(1,0.4,0.8)) )
}
```



$$W_i \sim \text{Distribution}(\mu_i, \dots)$$

$$\mu_i = k\pi p^2 H_i^3$$

$$p \sim \text{Beta}(25, 50)$$

$$k \sim \text{Exponential}(0.5)$$

*positive real,
variance scales with mean*

$$W_i \sim \text{LogNormal}(\mu_i, \sigma)$$

$$\exp(\mu_i) = k\pi p^2 H_i^3$$

$$p \sim \text{Beta}(25, 50)$$

$$k \sim \text{Exponential}(0.5)$$

$$\sigma \sim \text{Exponential}(1)$$

Growth is multiplicative,
log-normal is natural choice

*mu in log-normal is mean of log,
not mean of observed*

$$W_i \sim \text{LogNormal}(\mu_i, \sigma)$$

Growth is multiplicative,
log-normal is natural choice

$$\exp(\mu_i) = k\pi p^2 H_i^3$$

$$p \sim \text{Beta}(25, 50)$$

$$k \sim \text{Exponential}(0.5)$$

$$\sigma \sim \text{Exponential}(1)$$


```
## R code 16.2
```

```
dat <- list(W=d$w,H=d$h)
```

```
m16.1 <- ulam(
```

```
  alist(
```

```
    W ~ dlnorm( mu , sigma ),
```

```
    exp(mu) <- 3.141593 * k * p^2 * H^3,
```

```
    p ~ beta( 25 , 50 ),
```

```
    k ~ exponential( 0.5 ),
```

```
    sigma ~ exponential( 1 )
```

```
  ), data=dat , chains=4 , cores=4 )
```

$$W_i \sim \text{LogNormal}(\mu_i, \sigma)$$

$$\exp(\mu_i) = k\pi p^2 H_i^3$$

$$p \sim \text{Beta}(25, 50)$$

$$k \sim \text{Exponential}(0.5)$$

$$\sigma \sim \text{Exponential}(1)$$


```
## R code 16.2
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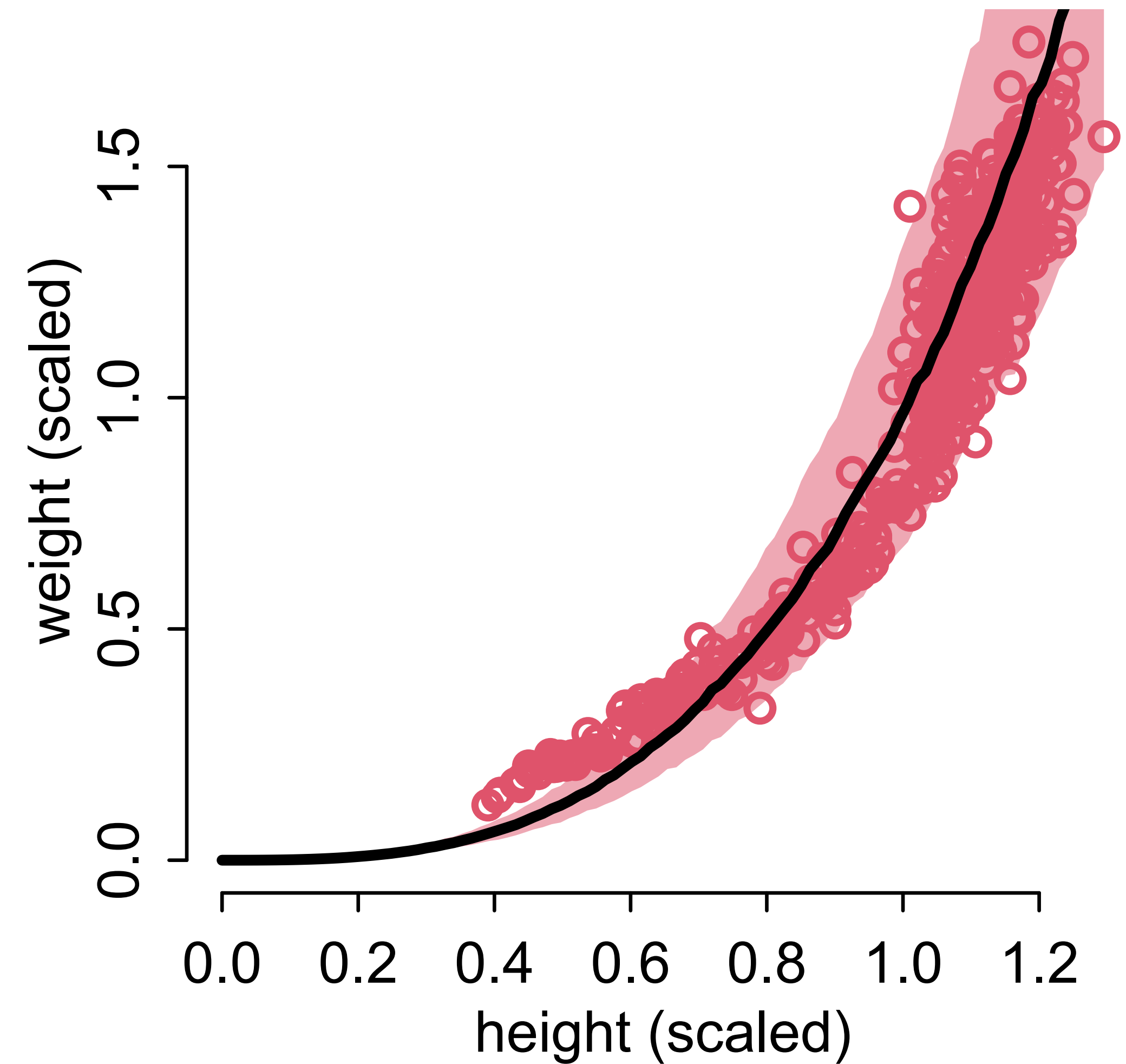
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    exp(mu) <- 3.141593 * k * p^2 * H^3,
```

```
    p ~ beta( 25 , 50 ),
```

```
    k ~ exponential( 0.5 ),
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```
    sigma ~ exponential( 1 )
```

```
  ), data=dat , chains=4 , cores=4 )
```



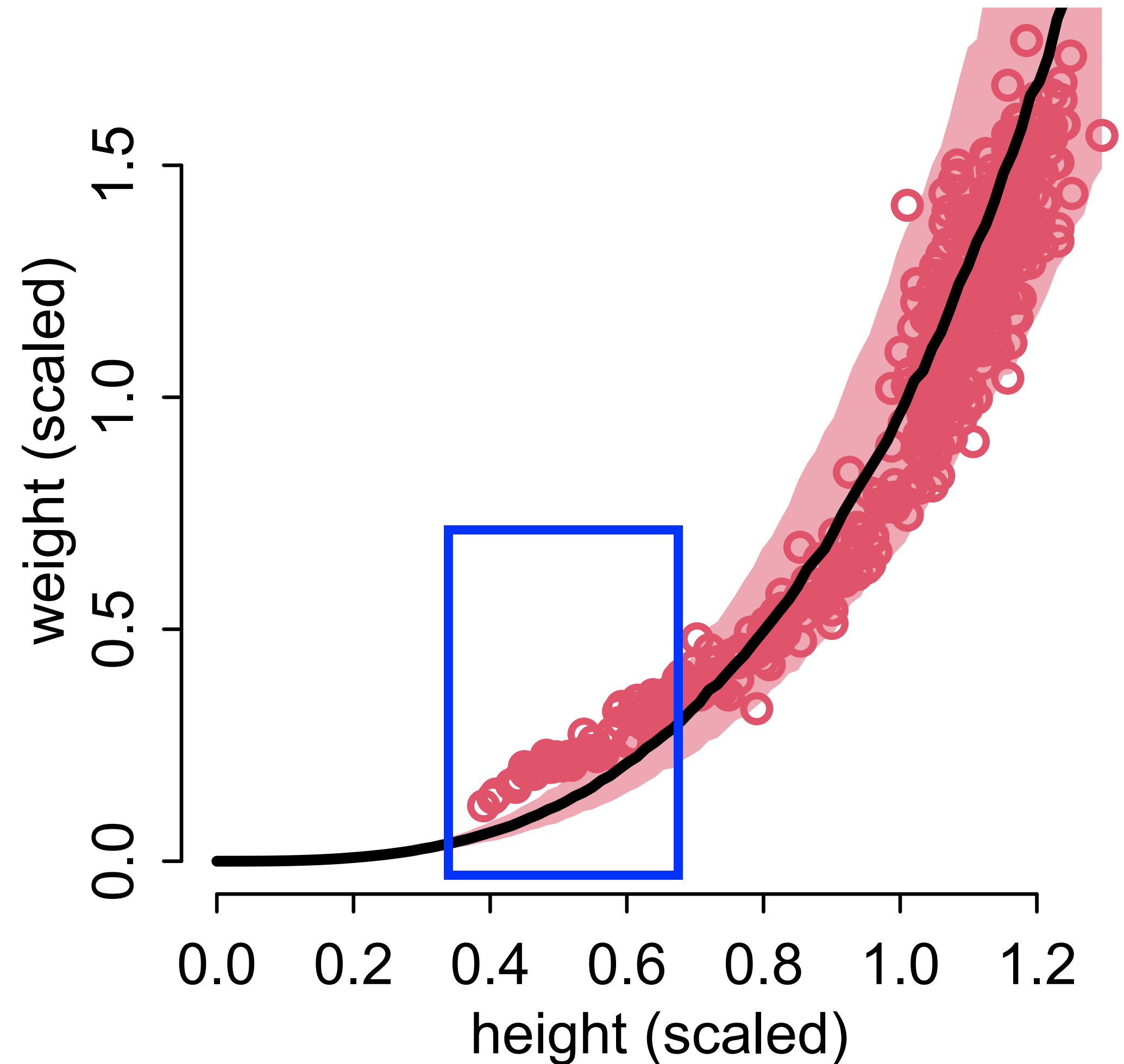
Insightful errors

Not bad for a cylinder

Poor fit for children

In scientific model, poor fit is
informative — p different for kids

Bad epicycles harder to read



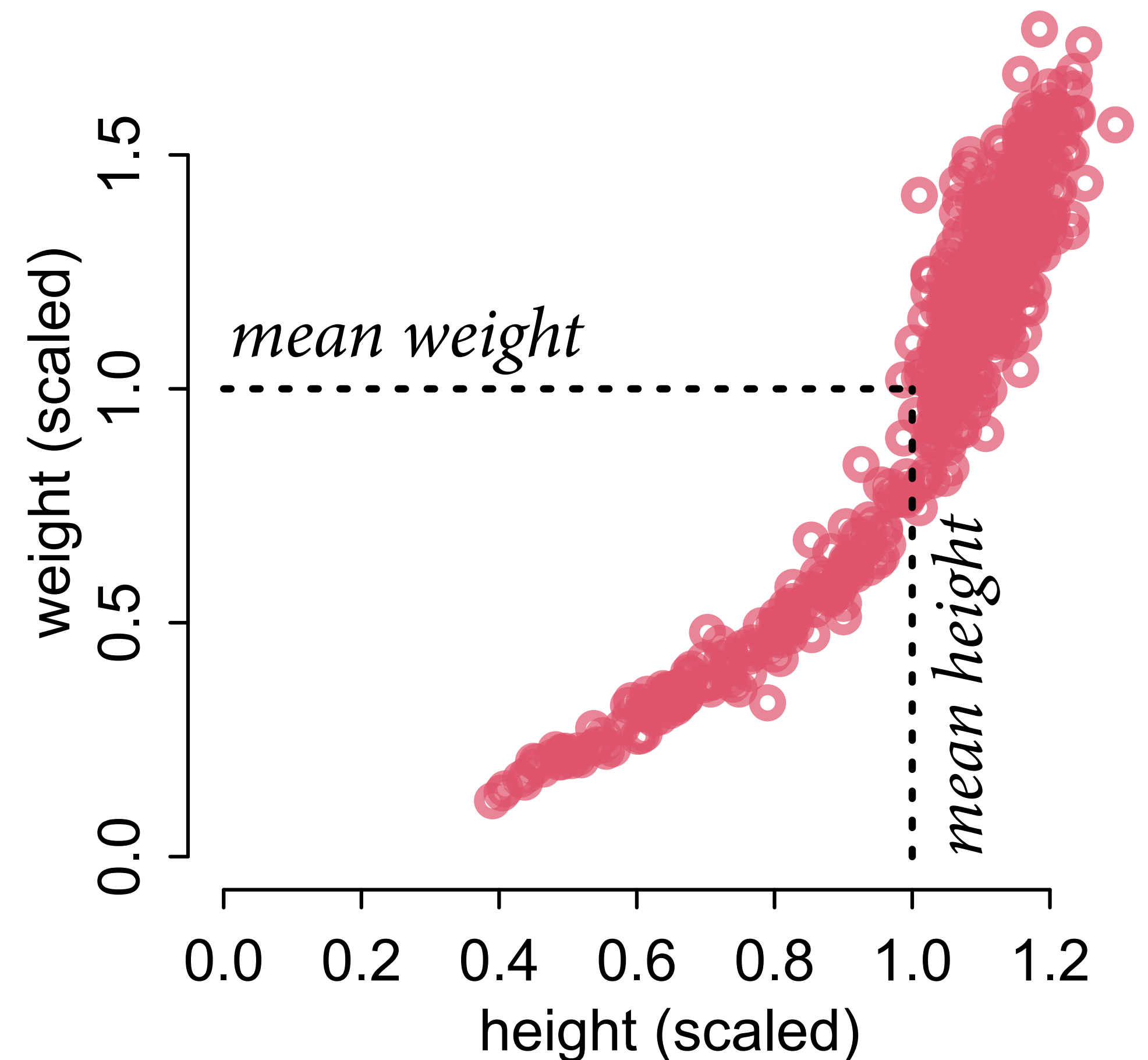
How to set these priors?

(1) Choose measurement scales

(2) Simulate

(3) Think

$$\mu_i = k\pi p^2 H_i^3$$



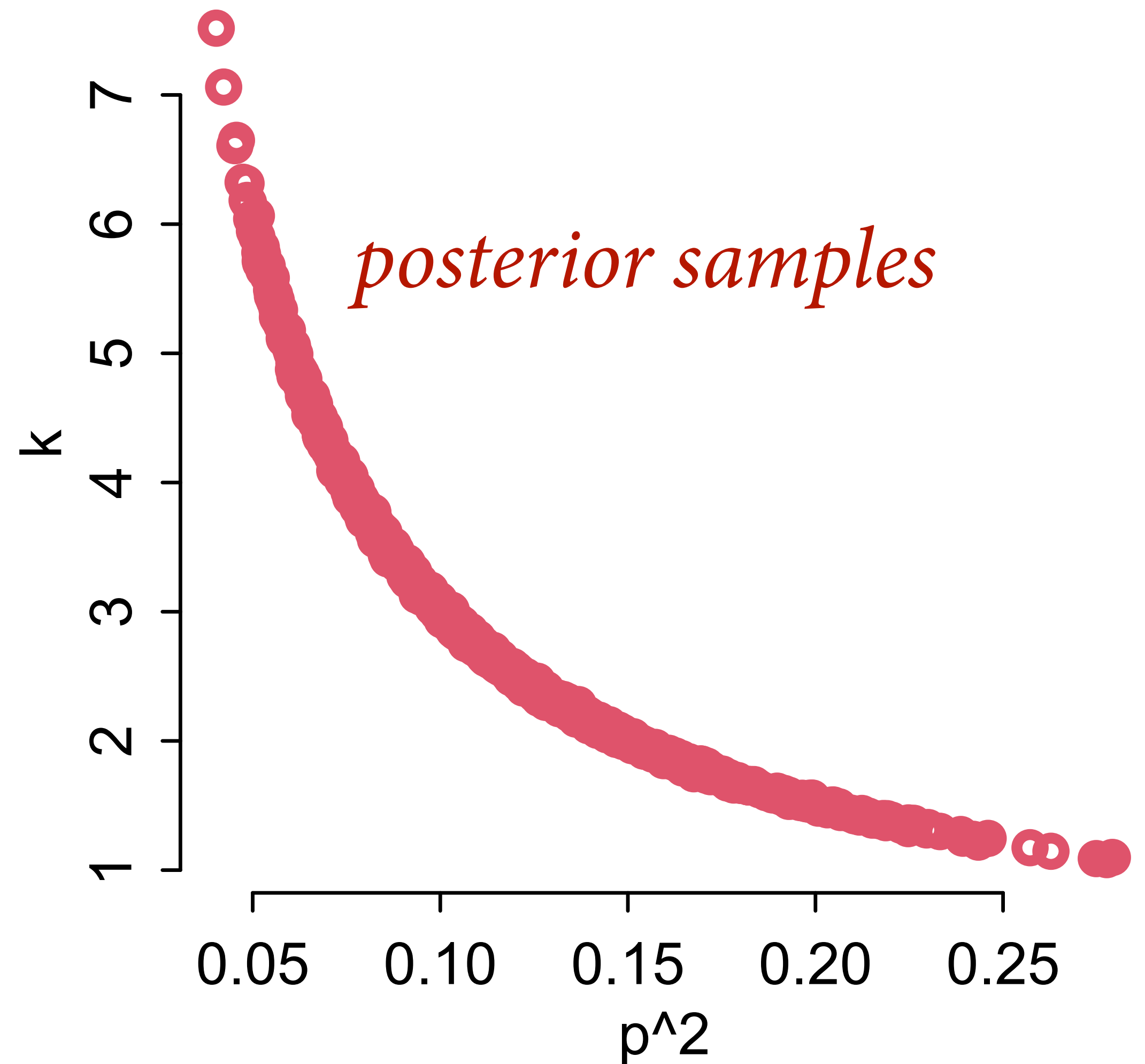
How to set these priors?

(1) Choose measurement scales

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$$\mu_i = k\pi p^2 H_i^3$$



How to set these priors?

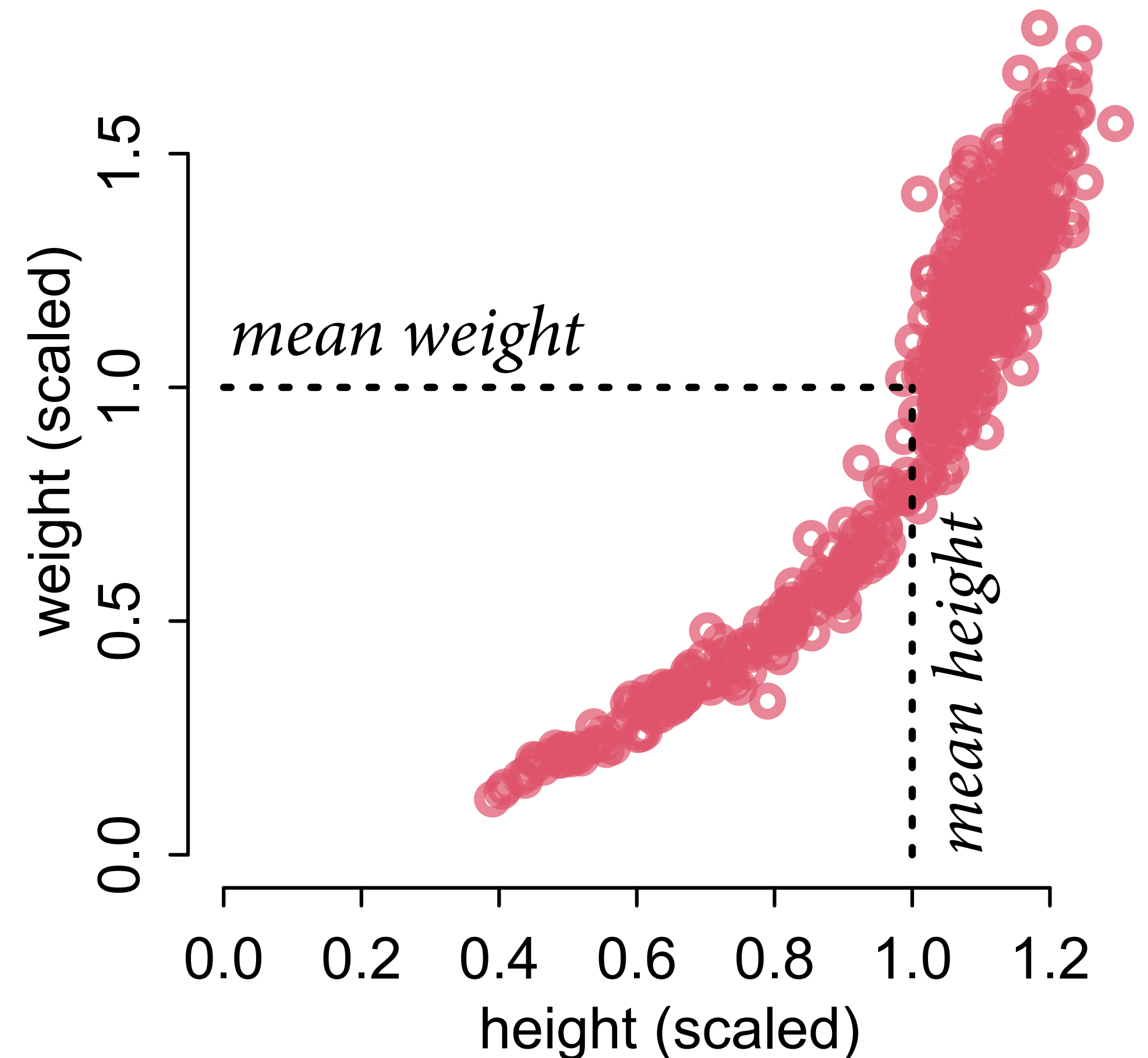
(1) Choose measurement scales

(2) Simulate

(3) Think

$$\mu_i = k\pi r^2 H_i^3$$

$$(1) = k\pi r^2 (1)^3$$



How to set these priors?

(1) Choose measurement scales

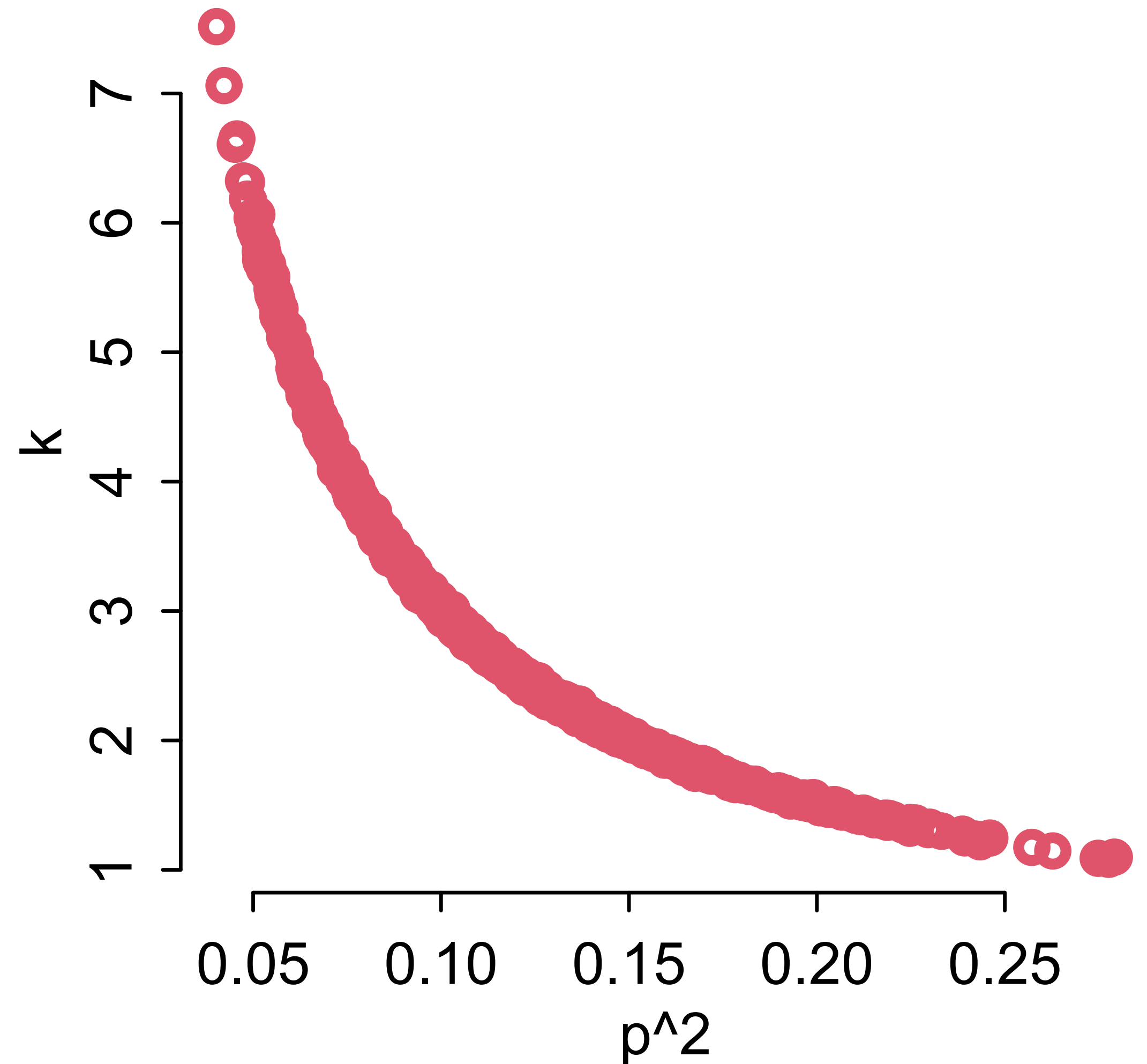
(2) Simulate

(3) Think

$$\mu_i = k\pi r^2 H_i^3$$

$$(1) = k\pi r^2 (1)^3$$

$$k = \frac{1}{\pi r^2}$$



How to set these priors?

(1) Choose measurement scales

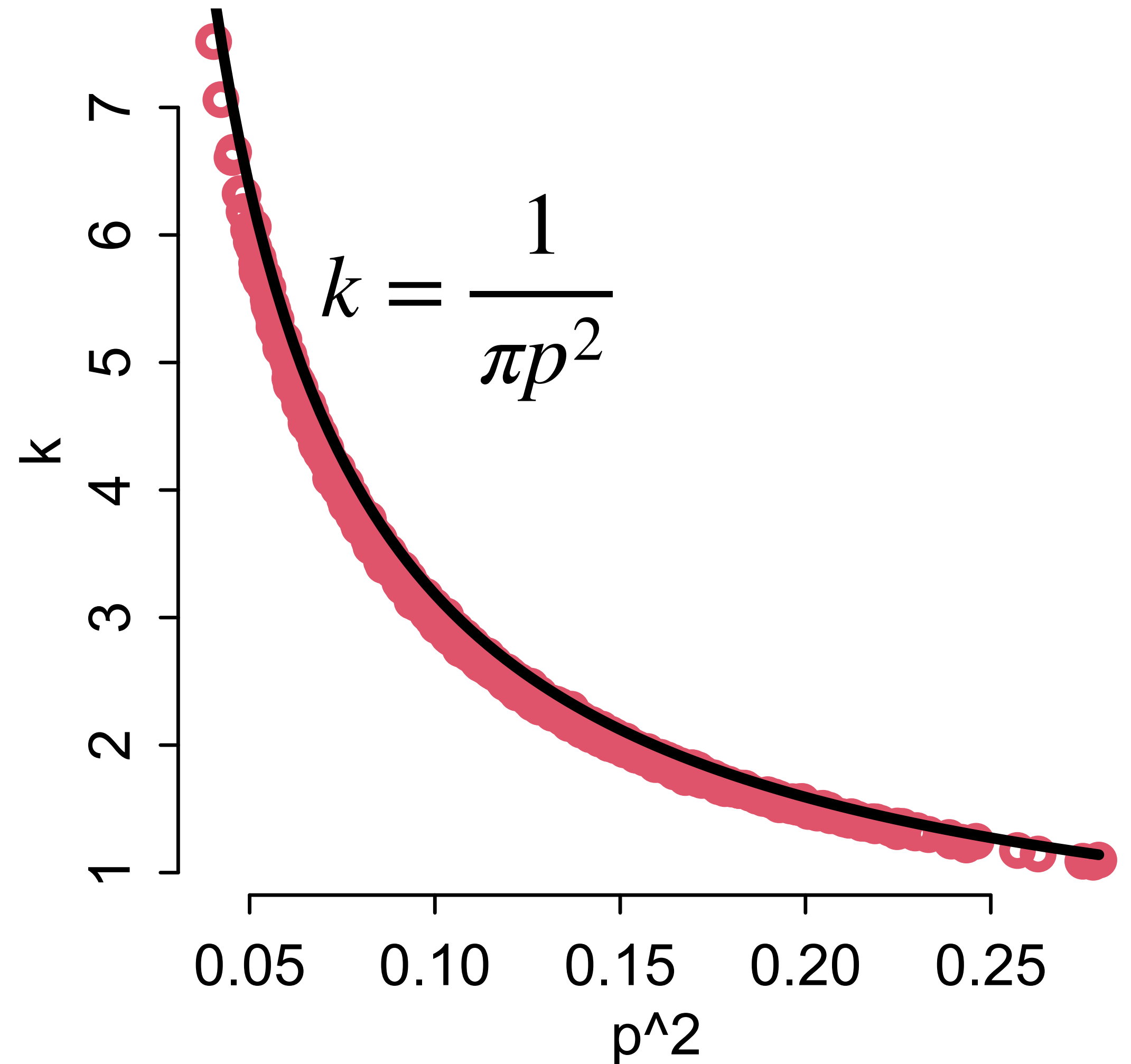
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$$\mu_i = k\pi r^2 H_i^3$$

$$(1) = k\pi r^2 (1)^3$$

$$k = \frac{1}{\pi r^2}$$



How to set these priors?

(1) Choose measurement scales

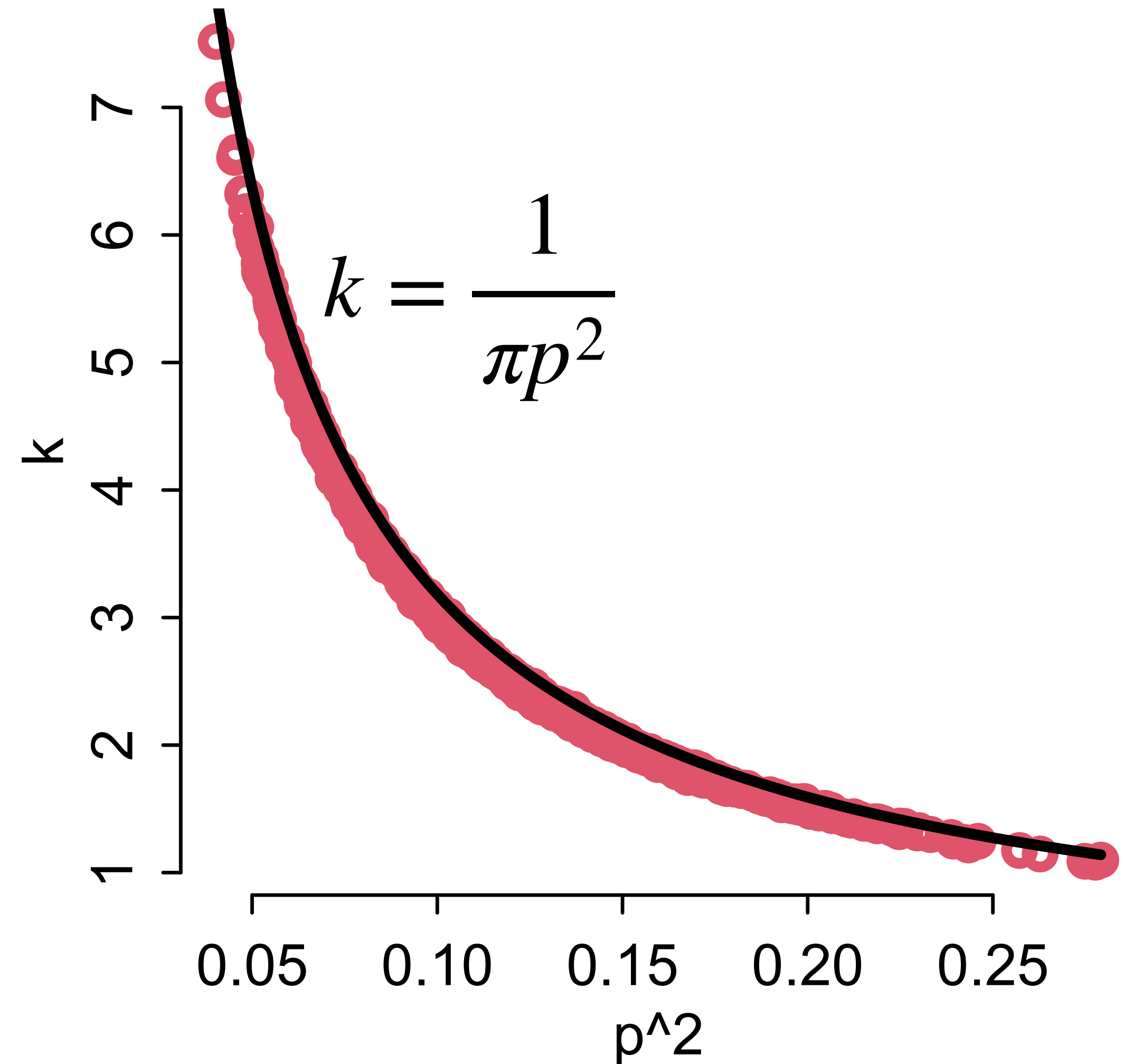
(2) Simulate

(3) Think

$$(1) = k\pi p^2(1)^3$$

$$(1) = \pi\theta(1)^3$$

$$\theta \approx \pi^{-1}$$



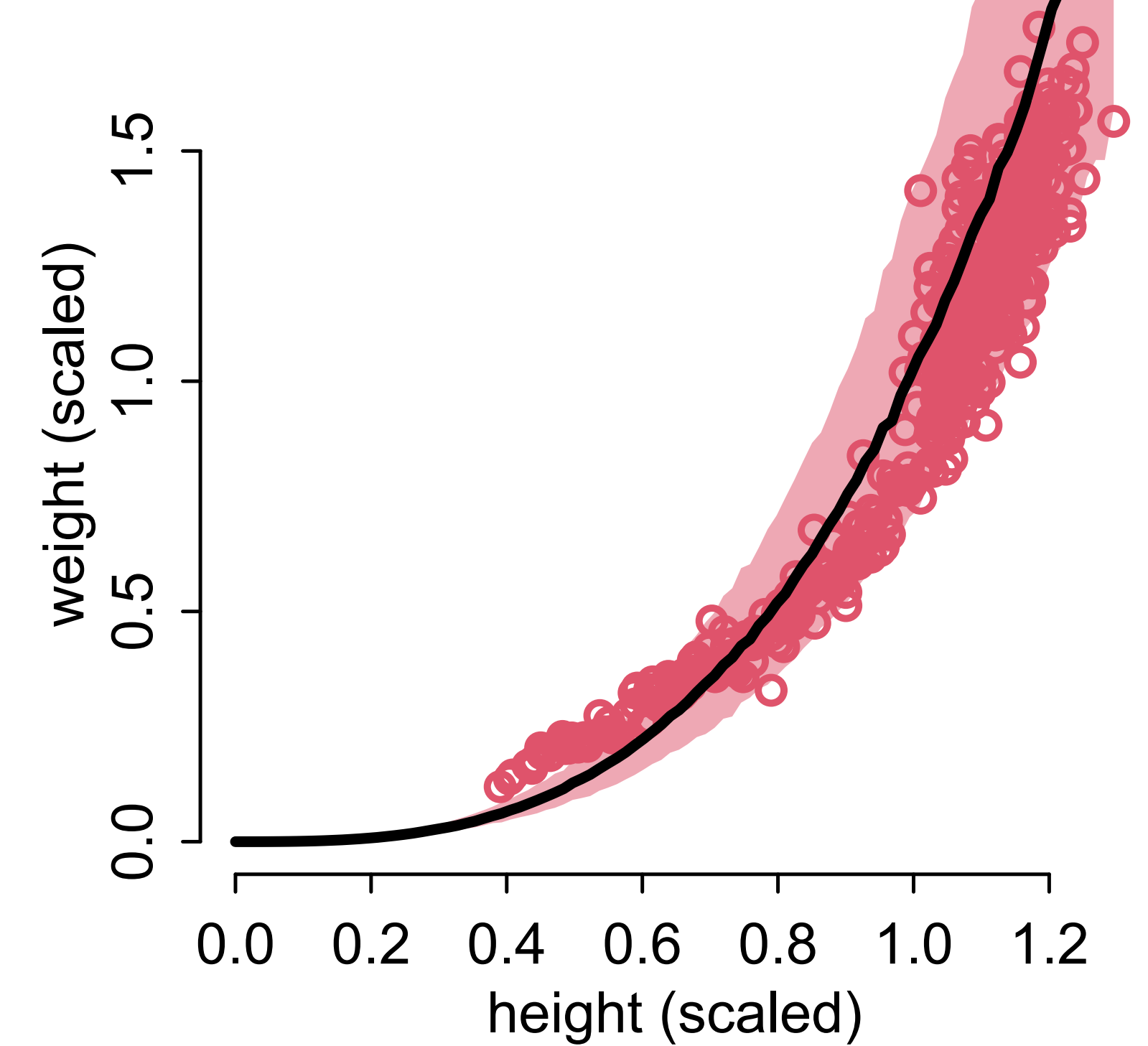

```
mWH2 <- ulam(  
  alist(  
    W ~ dlnorm( mu , sigma ),  
    exp(mu) <- H^3 ,  
    sigma ~ exponential( 1 )  
  ), data=dat , chains=4 , cores=4 )
```

In dimensionless model, W is H^3

$$W_i \sim \text{LogNormal}(\mu_i, \sigma)$$
$$\exp(\mu_i) = H_i^3$$
$$\sigma \sim \text{Exponential}(1)$$

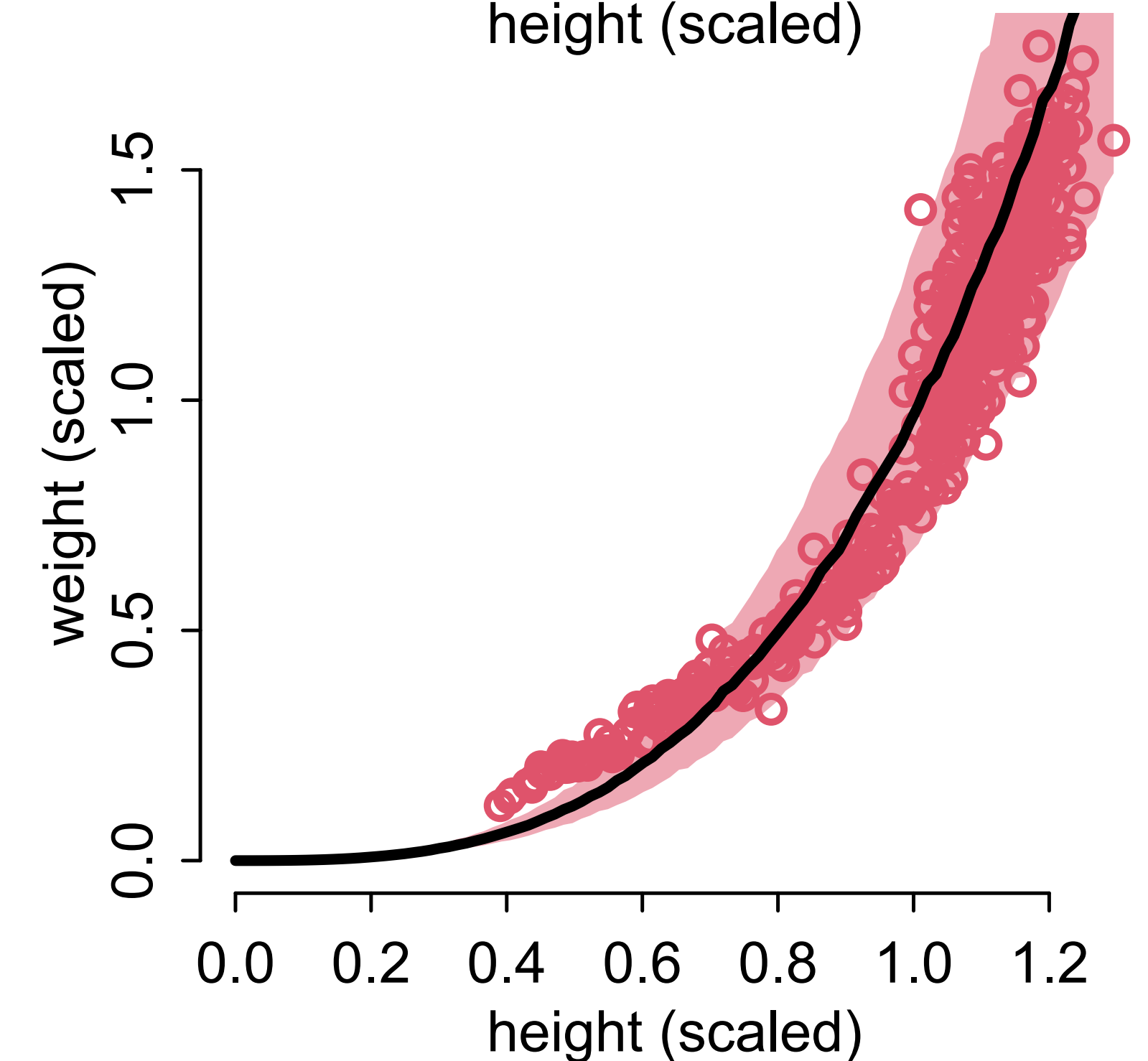
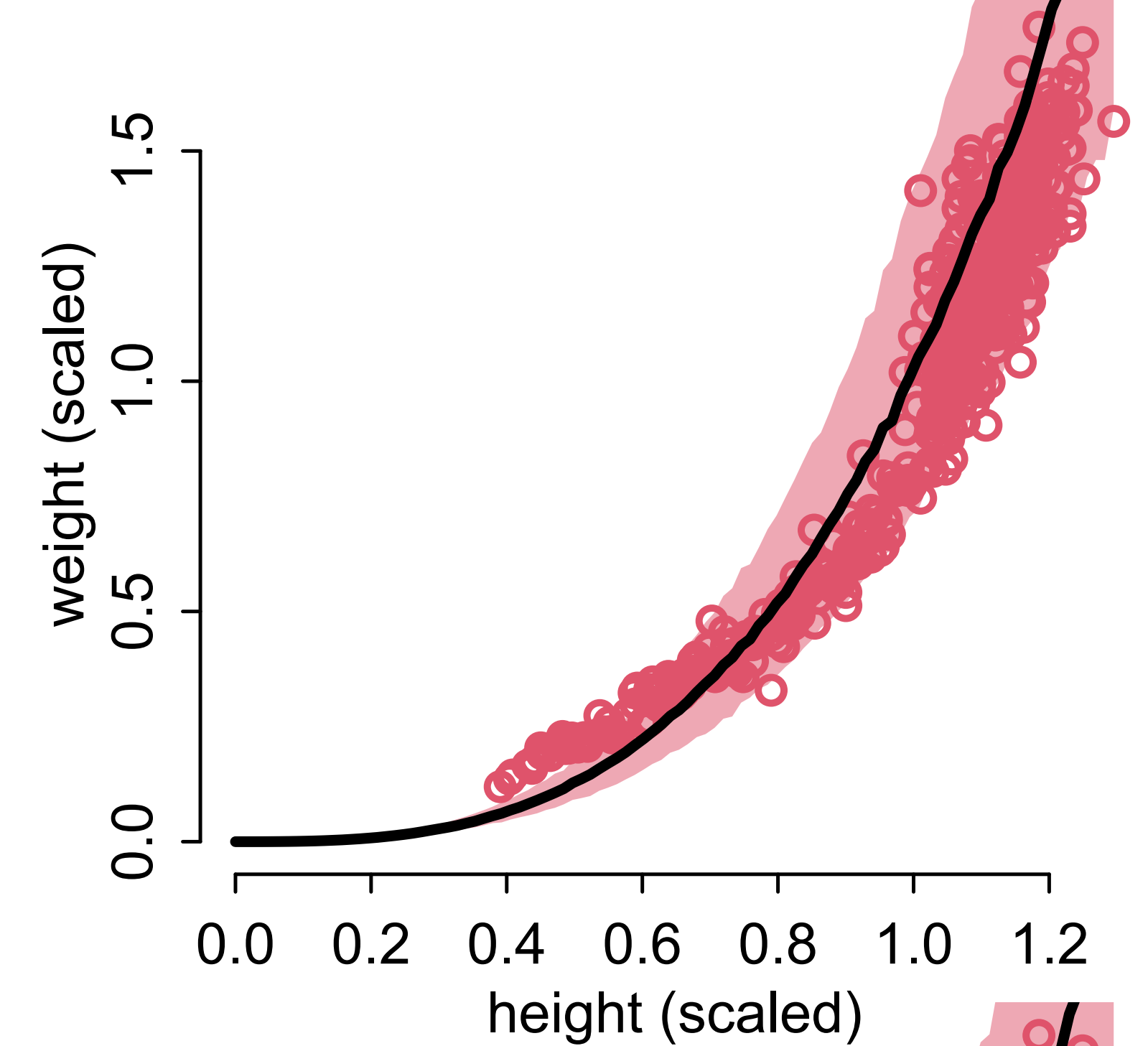
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In dimensionless model, W is H^3




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mWH2 <- ulam(
  alist(
    W ~ dlnorm( mu , sigma ),
    exp(mu) <- H^3 ,
    sigma ~ exponential( 1 )
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```

```
## R code 16.2
dat <- list(W=d$w,H=d$h)
m16.1 <- ulam(
  alist(
    W ~ dlnorm( mu , sigma ),
    exp(mu) <- 3.141593 * k * p^2 * H^3,
    p ~ beta( 25 , 50 ),
    k ~ exponential( 0.5 ),
    sigma ~ exponential( 1 )
  ), data=dat , chains=4 , cores=4 )
```



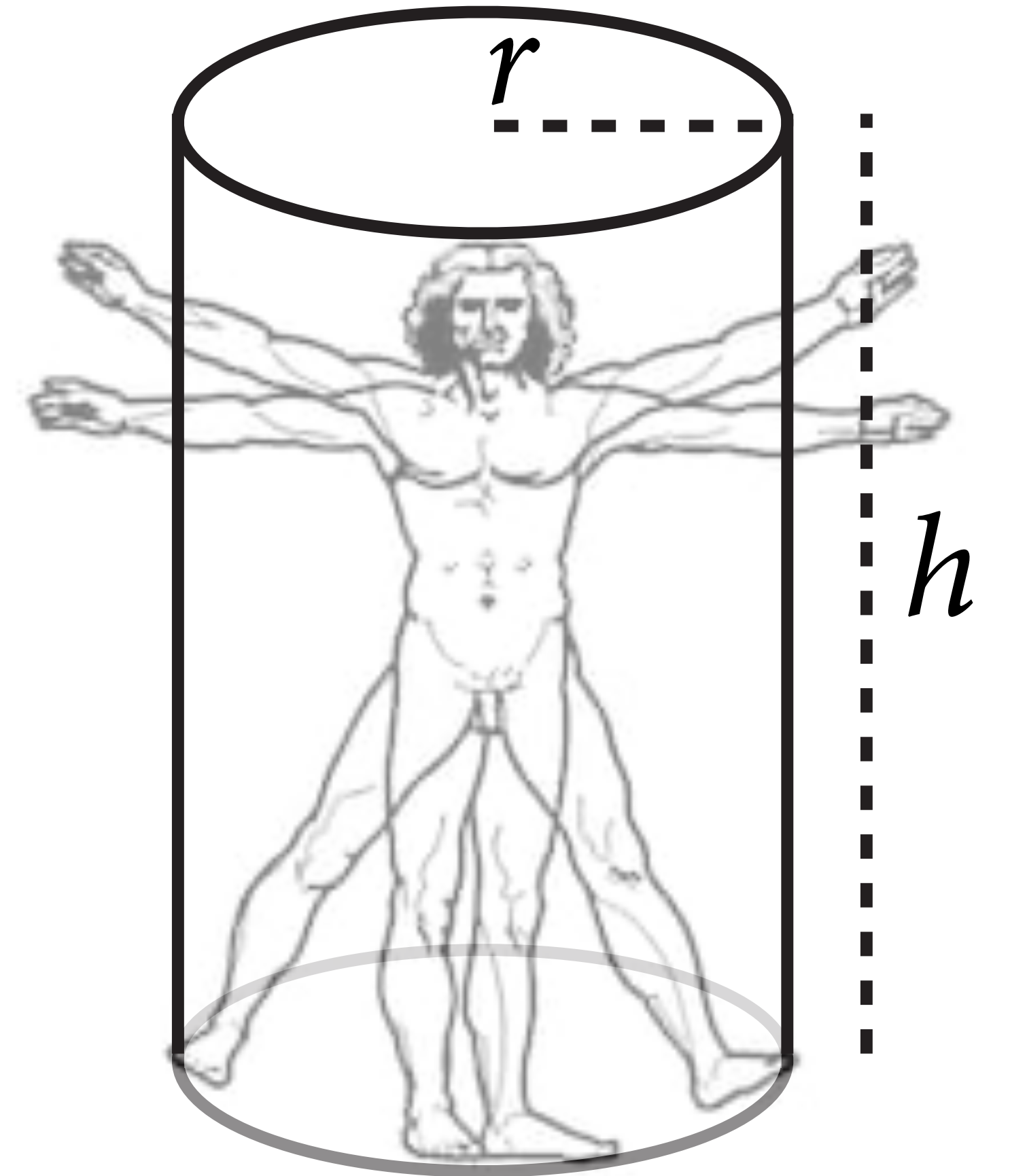
Geometric People

Most of the relationship $H \rightarrow W$ is just relationship between length and volume

Changes in body shape explain poor fit for children?

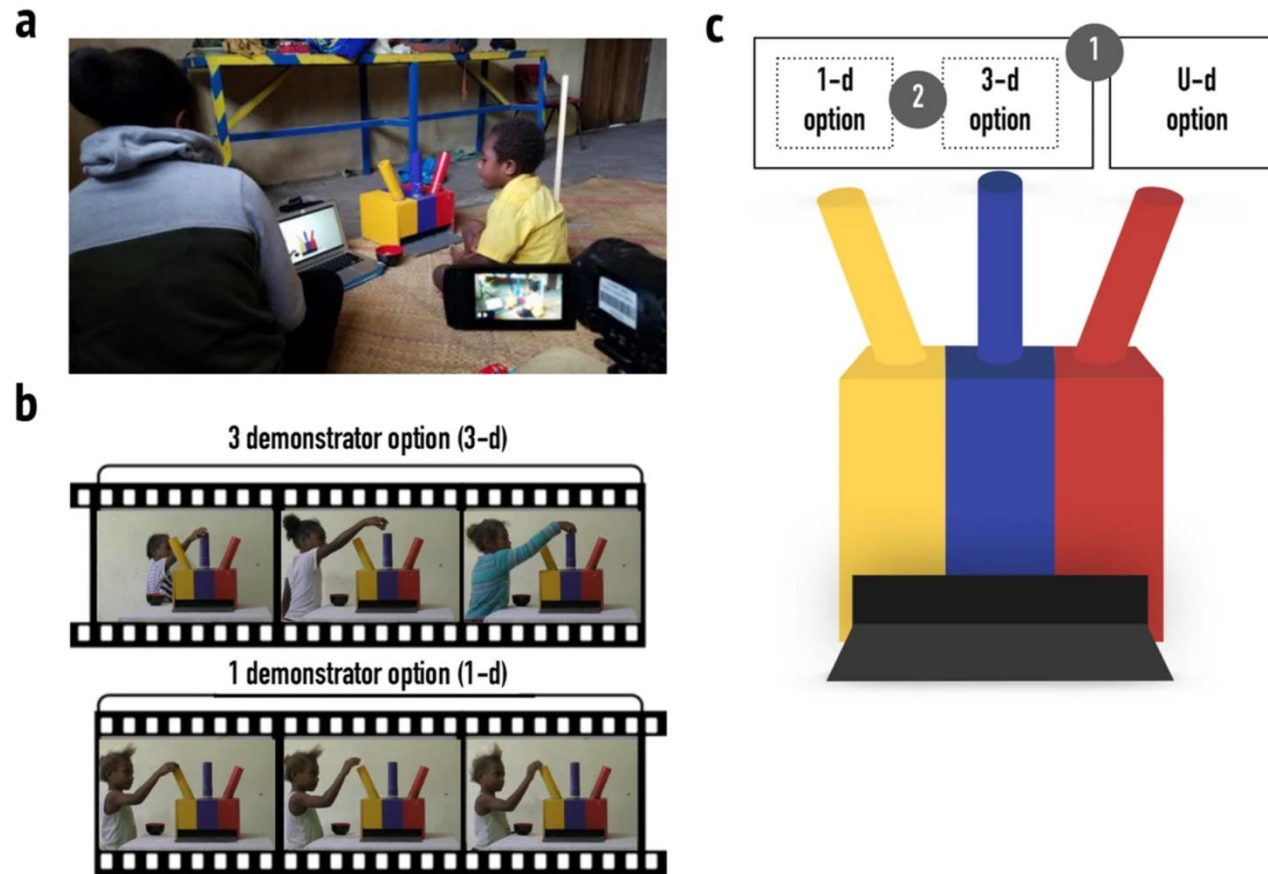
Problems provide insight when model is **scientific** instead of purely **statistical**

There is no empiricism without theory



$$W = k\pi r^2 h^3$$

Social learning and conformity



Social Conformity

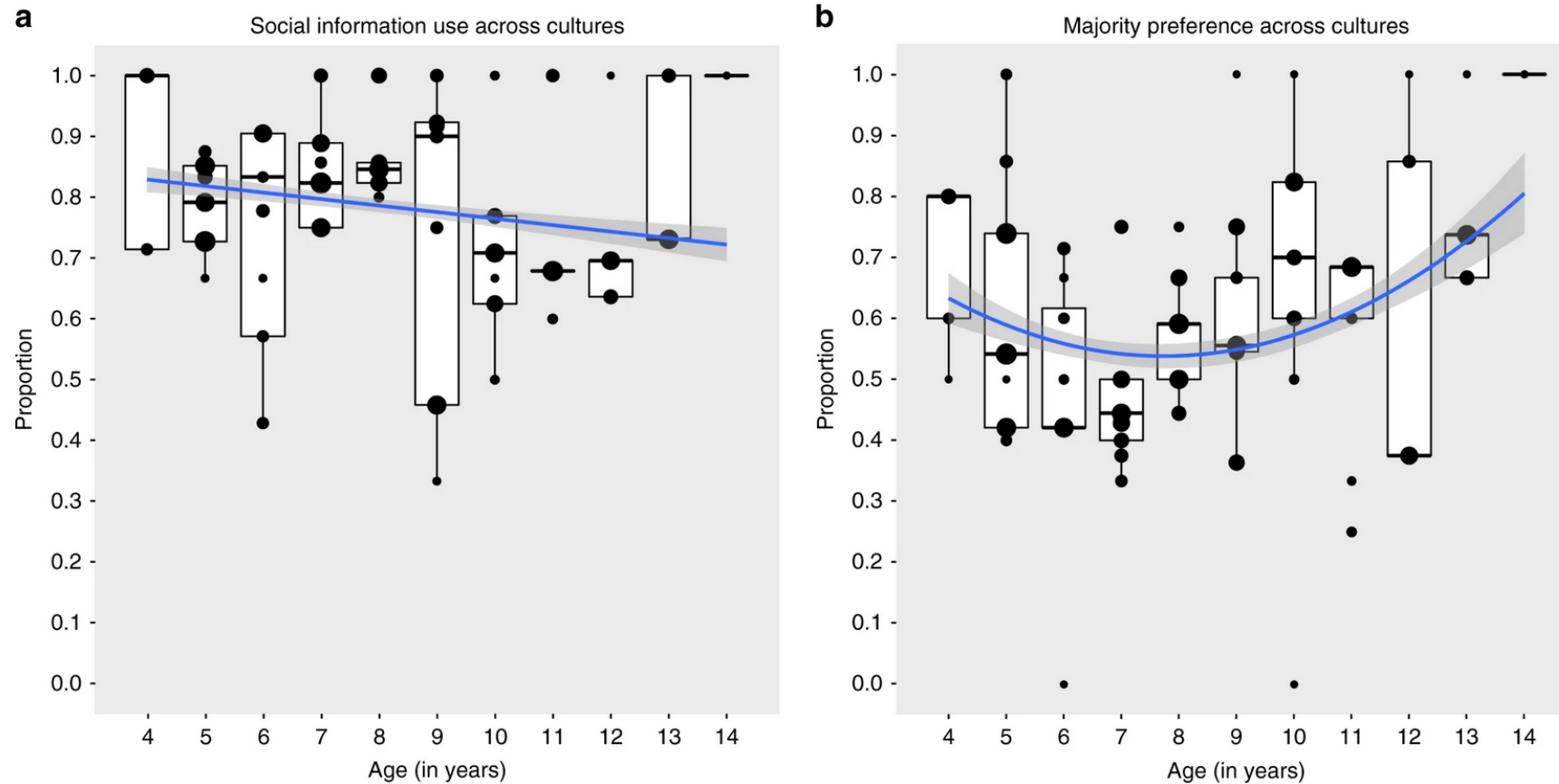
Do children copy the **majority**? If so, how does this develop?

Problem: Cannot see strategy, only choice

Majority choice consistent with many strategies



Originally published results



Social Conformity

Majority choice **consistent** with many strategies

Random color: Choose majority 1/3 of time

Random demonstrator: 3/4 of time

Random demonstration: 1/2 of time

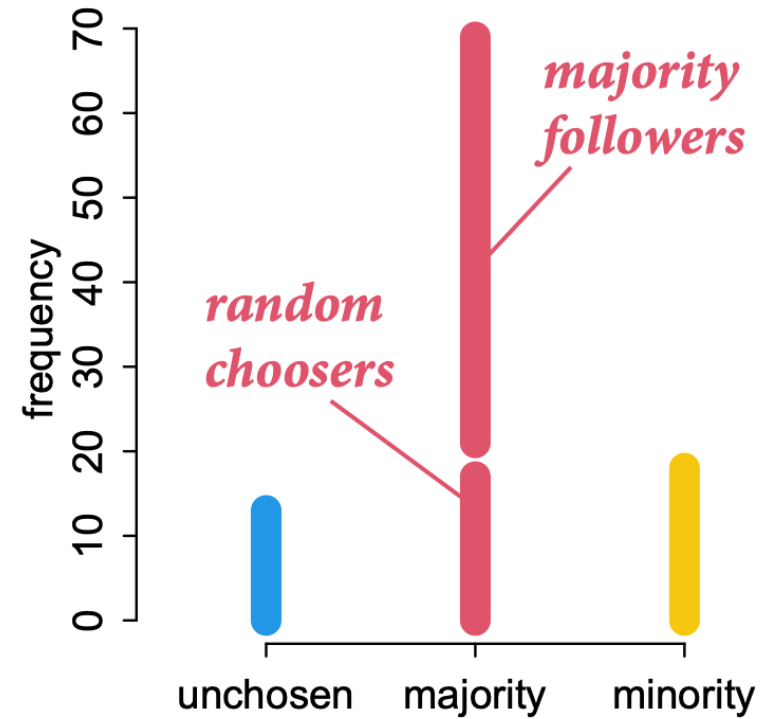


```
N <- 100 # number of children

# half choose random color
# sample from 1,2,3 at random for each
y1 <- sample( 1:3 , size=N/2 , replace=TRUE )

# half follow majority
y2 <- rep( 2 , N/2 )

# combine and shuffle y1 and y2
y <- sample( c(y1,y2) )
```



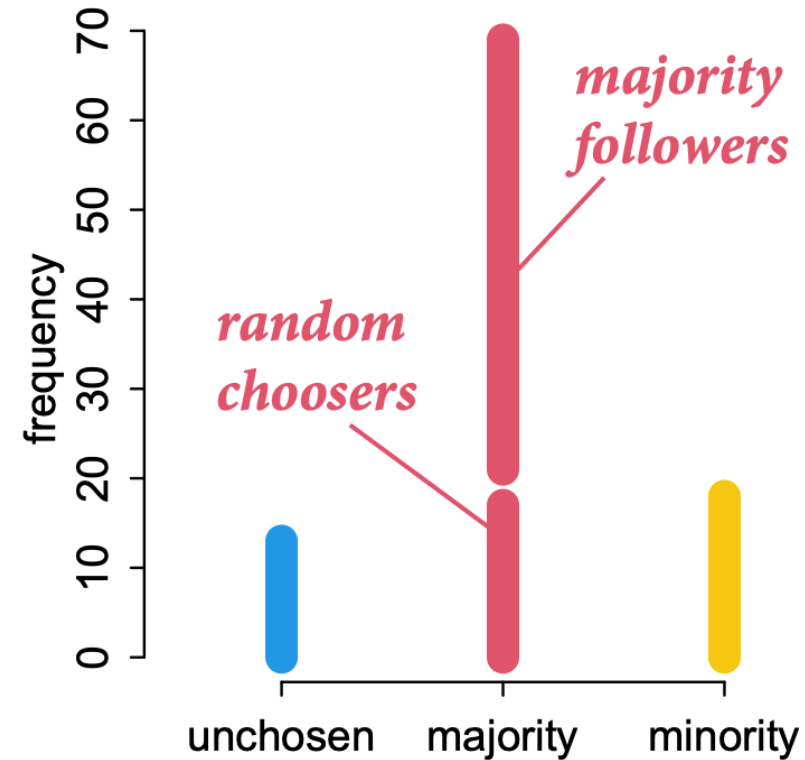
State-Based Model

Majority choice does not indicate majority preference

Instead infer the unobserved strategy (state) of each child

Strategy space:

- (1) **Majority** (2) **Minority**
- (3) **Maverick** (4) Random Color
- (5) Follow First



$$Y_i \sim \text{Categorical}(\theta)$$

*vector with probability
of each choice*

*Probability of (1) unchosen,
(2) majority, (3) minority*

$$Y_i \sim \text{Categorical}(\theta)$$

*Probability of (1) unchosen,
(2) majority, (3) minority*

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

Probability of choice j

average over strategies

prior probability strategy S

probability choice j assuming strategy S

$$Y_i \sim \text{Categorical}(\theta)$$

*Probability of (1) unchosen,
(2) majority, (3) minority*

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

$$p \sim \text{Dirichlet}([4,4,4,4,4])$$

Prior for strategy space

Stan code instead of Ulam this time

```
data{
  int N;
  int y[N];
  int majority_first[N];
}
parameters{
  simplex[5] p;
}
model{
  vector[5] theta_j;

  // prior
  p ~ dirichlet( rep_vector(4,5) );

  // probability of data
  for ( i in 1:N ) {
    theta_j = rep_vector(0,5); // clear it out
    if ( y[i]==2 ) theta_j[1]=1; // majority
    if ( y[i]==3 ) theta_j[2]=1; // minority
    if ( y[i]==1 ) theta_j[3]=1; // maverick
    theta_j[4]=1.0/3.0; // random color
    if ( majority_first[i]==1 ) // follow first
      if ( y[i]==2 ) theta_j[5]=1;
    else
      if ( y[i]==3 ) theta_j[5]=1;
  }
}
```

$Y_i \sim \text{Categorical}(\theta)$

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

$p \sim \text{Dirichlet}([4,4,4,4,4])$

```

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    if ( majority_first[i]==1 ) // follow first
      if ( y[i]==2 ) theta_j[5]=1;
    else
      if ( y[i]==3 ) theta_j[5]=1;

    // compute log( p_S * Pr(y_i|S) )
    for ( S in 1:5 )
      theta_j[S] = log(p[S]) + log(theta_j[S]);

    // compute average log-probability of y_i
    target += log_sum_exp( theta_j );
  }
}

```

$Y_i \sim \text{Categorical}(\theta)$

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

$p \sim \text{Dirichlet}([4,4,4,4,4])$

```

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```

$Y_i \sim \text{Categorical}(\theta)$

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

$p \sim \text{Dirichlet}([4,4,4,4,4])$

```

model{
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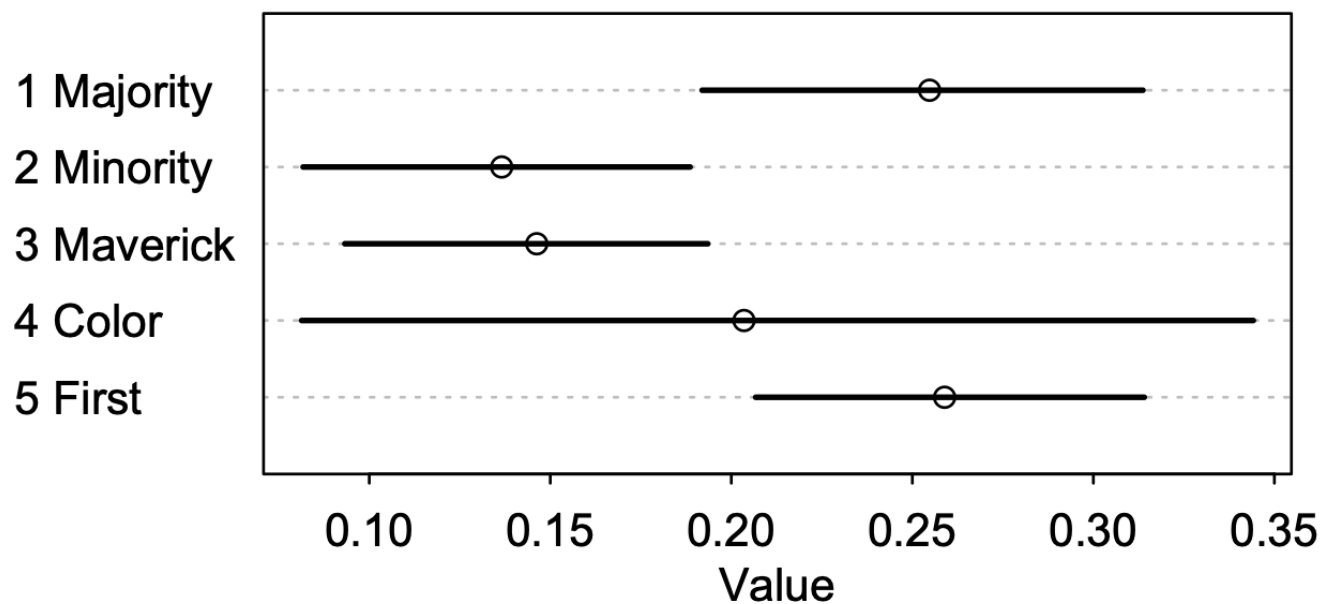
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  }
}

```

$Y_i \sim \text{Categorical}(\theta)$

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

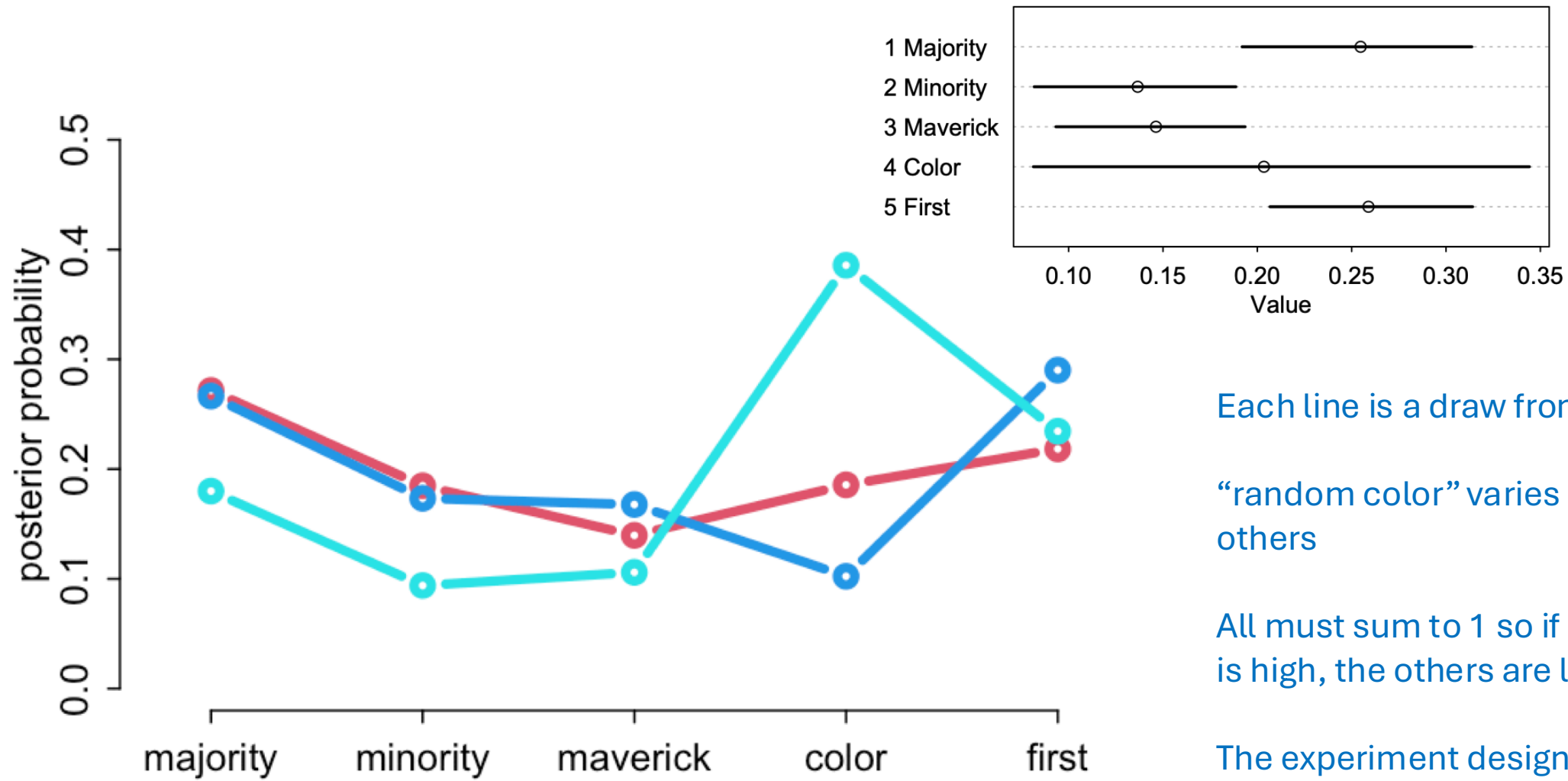
$p \sim \text{Dirichlet}([4,4,4,4,4])$



$$Y_i \sim \text{Categorical}(\theta)$$

$$\theta_j = \sum_{S=1}^5 p_S \Pr(Y = j | S)$$

$$p \sim \text{Dirichlet}([4,4,4,4,4])$$



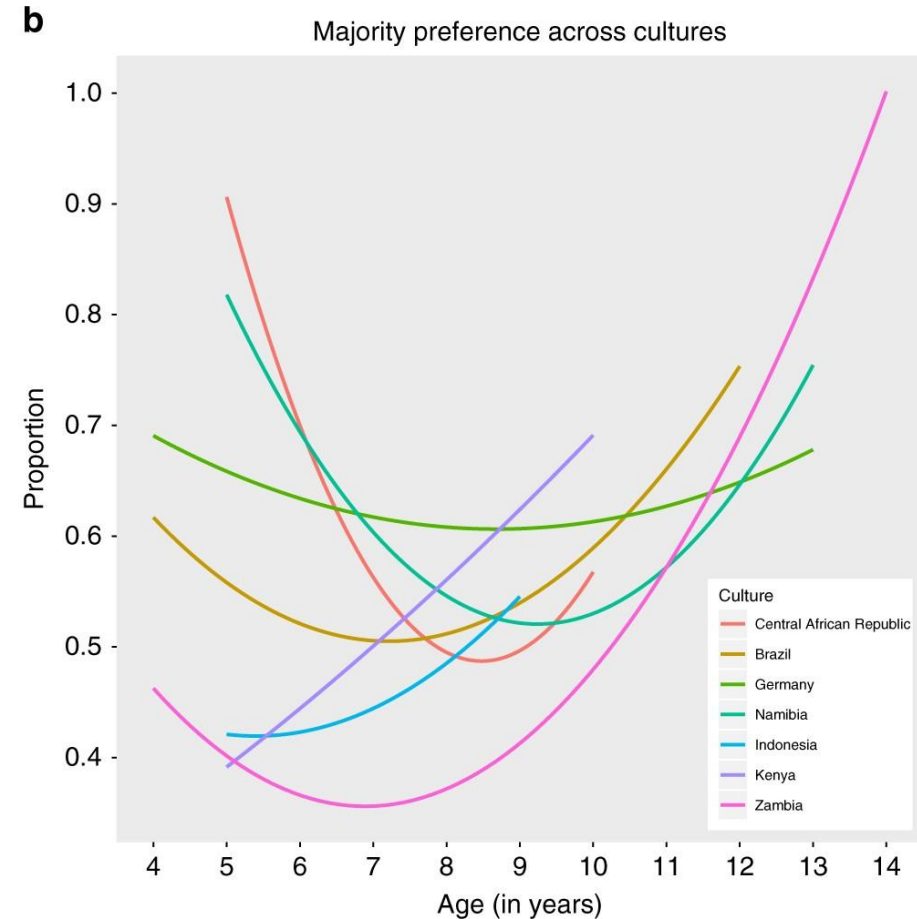
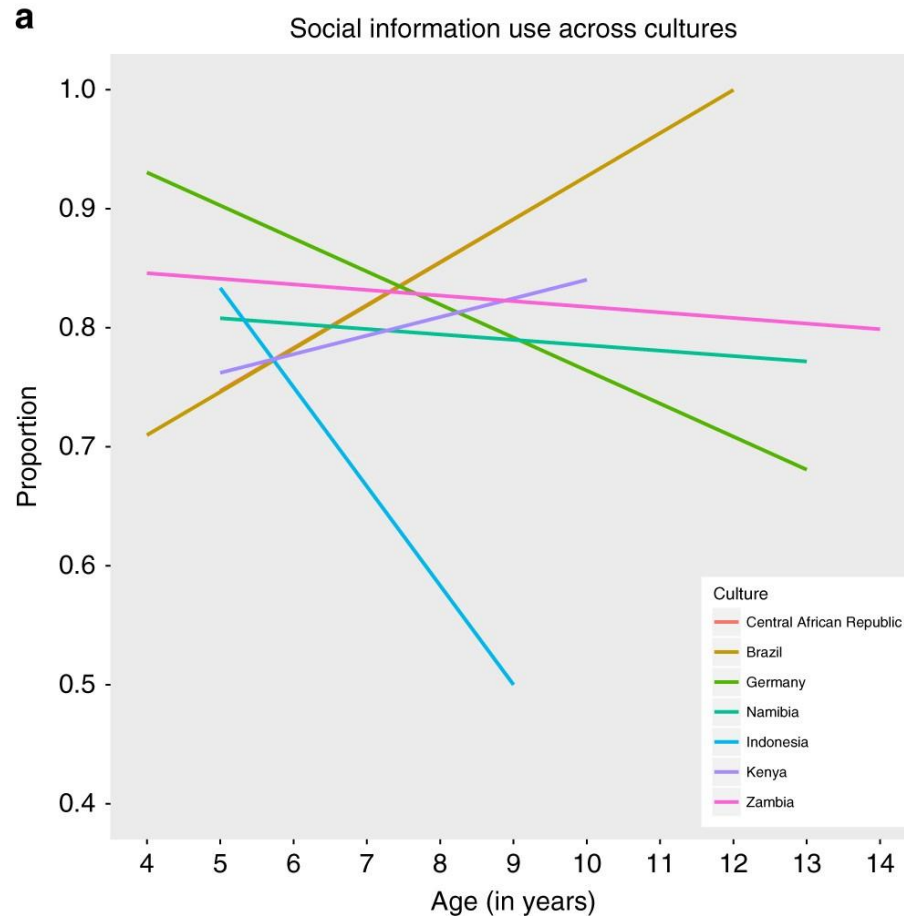
Each line is a draw from the posterior

“random color” varies more than the others

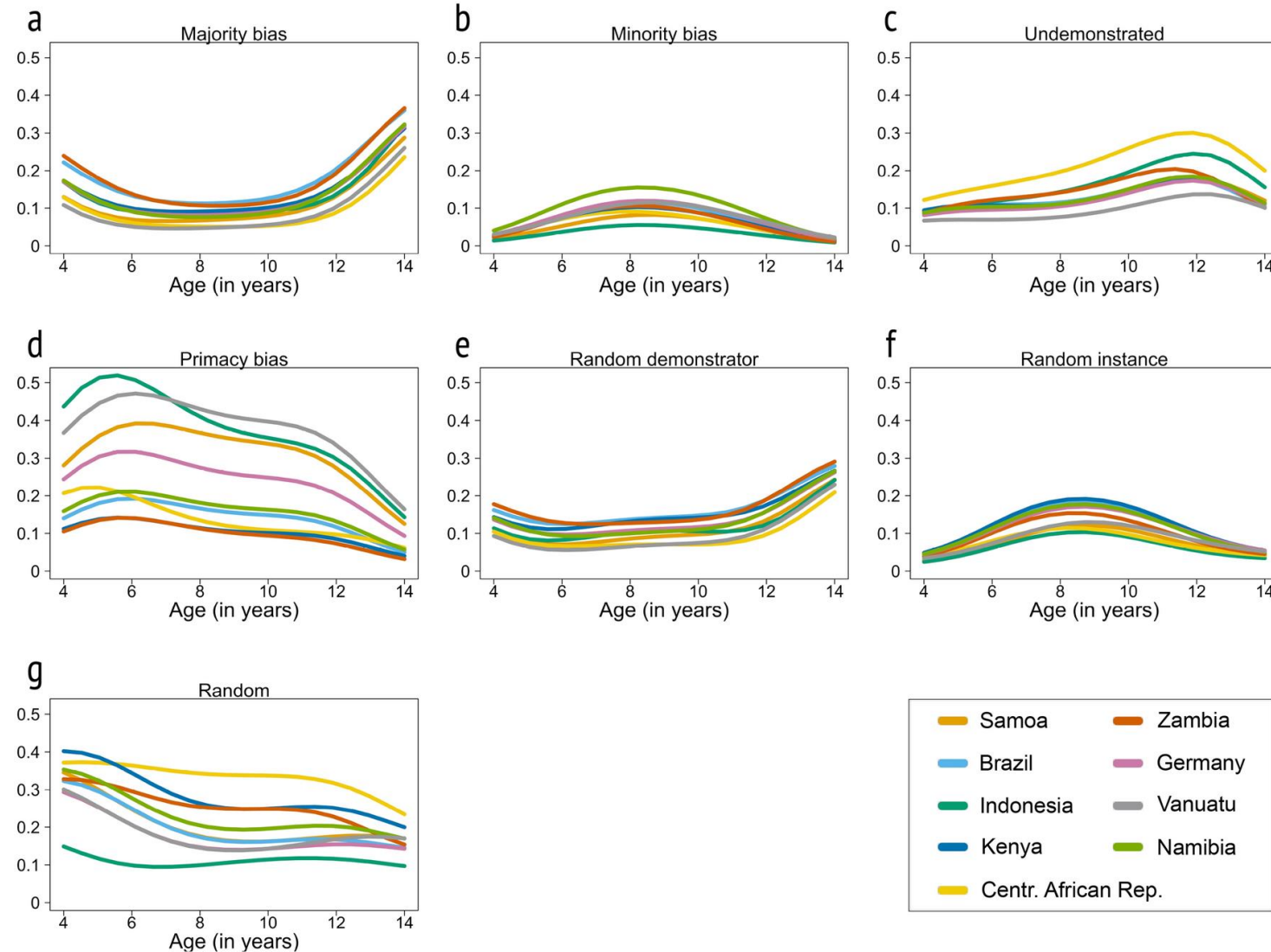
All must sum to 1 so if random color is high, the others are lower

The experiment design can’t exclude random choice

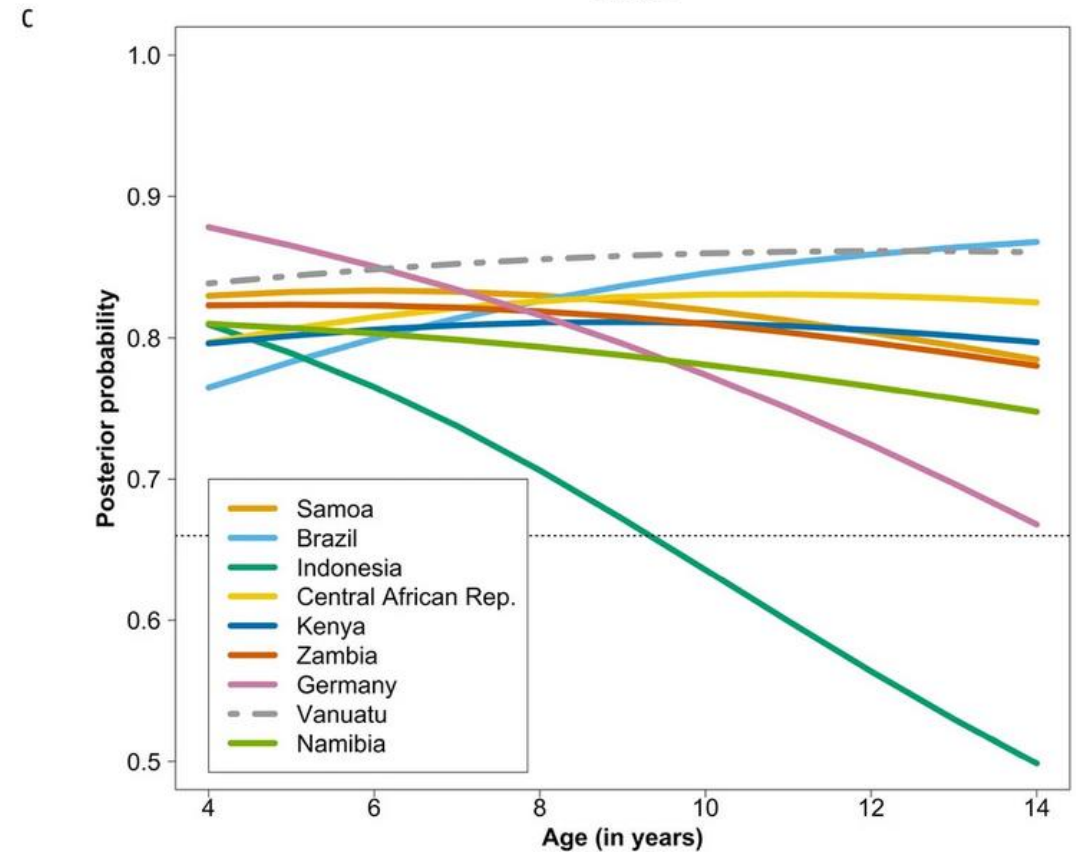
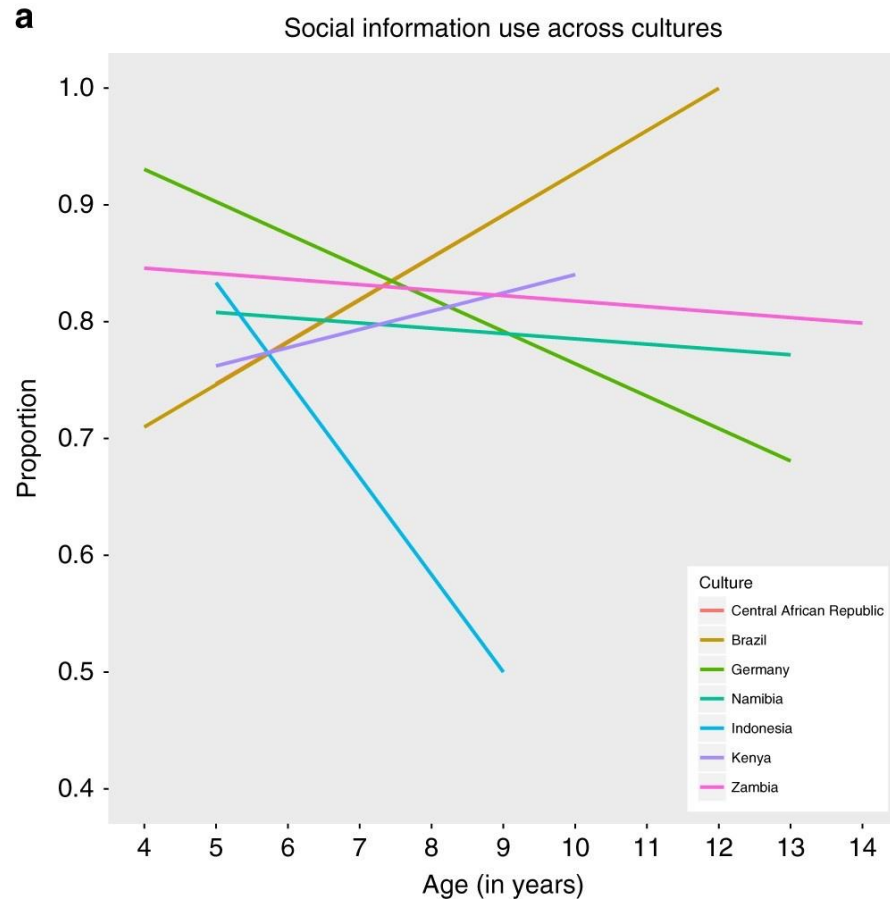
van Leeuwen et al 2018 Nat Comms

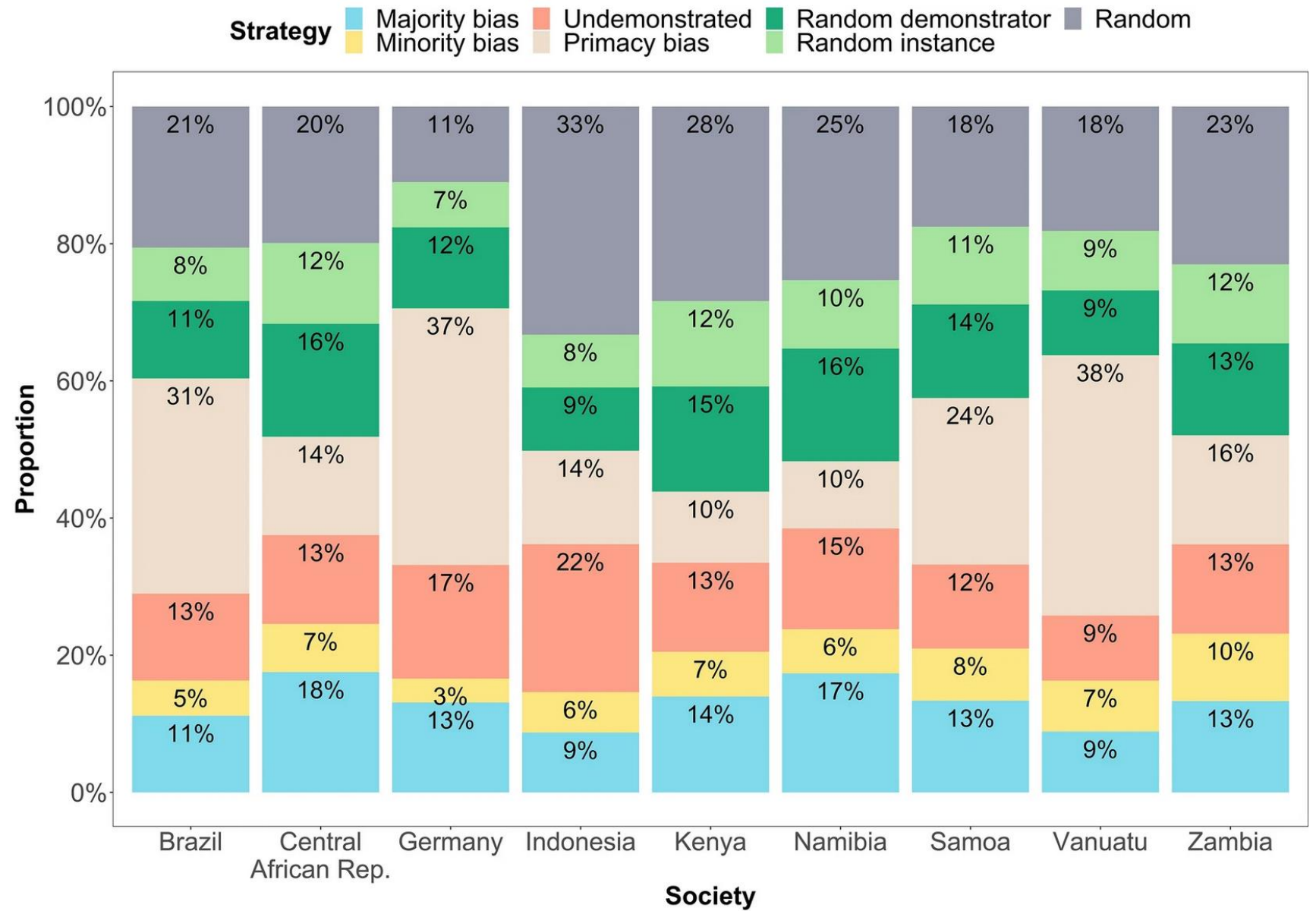


Sibilsky et al 2022 Sci Reports



Social information use across age





Inferred posterior mean probabilities of the seven selected learning strategies by society averaged over age.

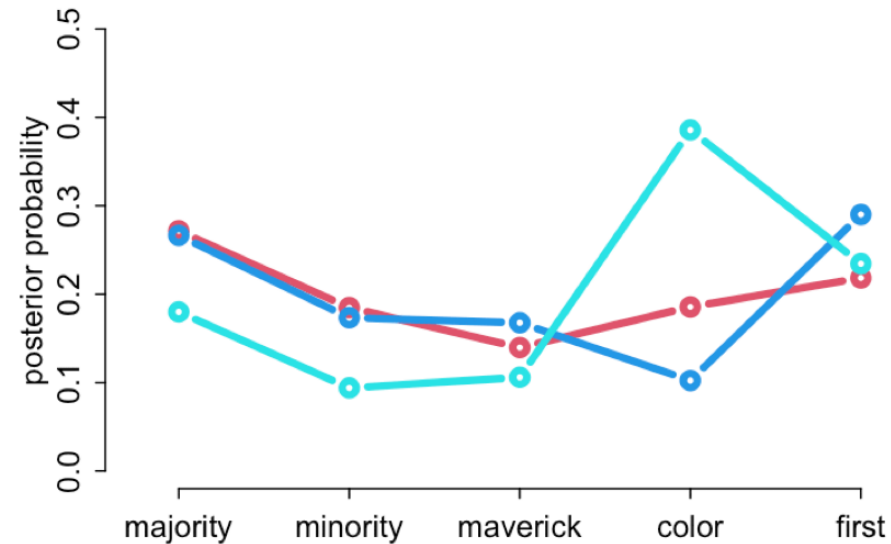
State-Based Models

What we want: Latent states

What we have: Emissions

Typically lots of uncertainty, but
being honest is only ethical choice

Large family: Movement, learning,
population dynamics, international
relations, family planning, ...



Population Dynamics

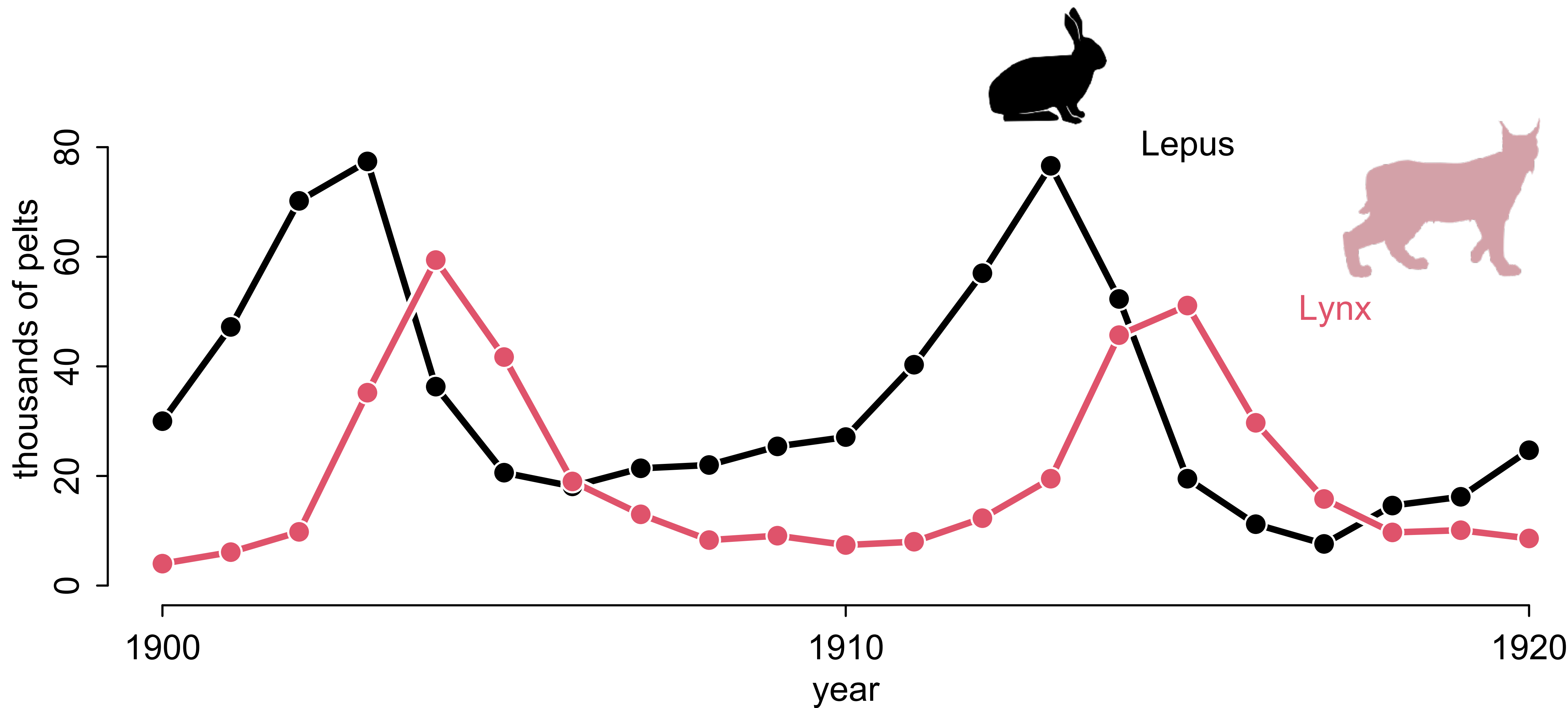
Latent states can be time varying

Example: Ecological dynamics,
numbers of different species over
time

Estimand: How do different species
interact; how do interactions
influence population dynamics

How to Draw a **Lynx**





recorded by the Hudson Bay Co.

A common geocentric way to model a time series like this would be to use something called an **AUTOREGRESSIVE MODEL**. In an autoregressive model, the value of the outcome in the previous time step is called a *lag variable* and added as a predictor to the right side of a linear model. For example, we might model the mean number of hares at time t as:

$$E(H_t) = \alpha + \beta_1 H_{t-1}$$

where H_t is the number of hares at time t . If β_1 is less than 1, then hares tend to regress to some mean population size α . We could continue by adding an epicycle for the predator:

$$E(H_t) = \alpha + \beta_1 H_{t-1} + \beta_2 L_{t-1}$$

where L_{t-1} is the number of lynx in the previous time period. Sometimes people add even deeper lags, like this:

$$E(H_t) = \alpha + \beta_1 H_{t-1} + \beta_2 L_{t-1} + \beta_3 H_{t-2}$$

Now not only does the most recent population size H_{t-1} predict the present, but so too does the population size two time periods ago H_{t-2} . Everything from prices to temperature to wars has been modeled this way.

Issues with autoregressive models

- No lag beyond one time period makes any causal sense
 - Nothing that happened 2+ periods ago causes the present state except through an influence on the intervening state(s)
 - Some causes may act more slowly than others
 - Need another variable in this case, not additional lags

Issues with autoregressive models

- If states are measured with error, then the model propagates error without accounting for it.
 - E.g., the observed H_{t-1} does not influence H_t , rather the unobserved true H_{t-1} does.

Issues with autoregressive models

- In most cases there is no biological, economic, or physical interpretation of the parameters.

Consider for example the α intercept in the equation,

$$E(H_t) = \alpha + \beta_1 H_{t-1}$$

It implies that even when there are no hares, $H_{t-1} = 0$, there can be α hares in the next period.

Issues with autoregressive models

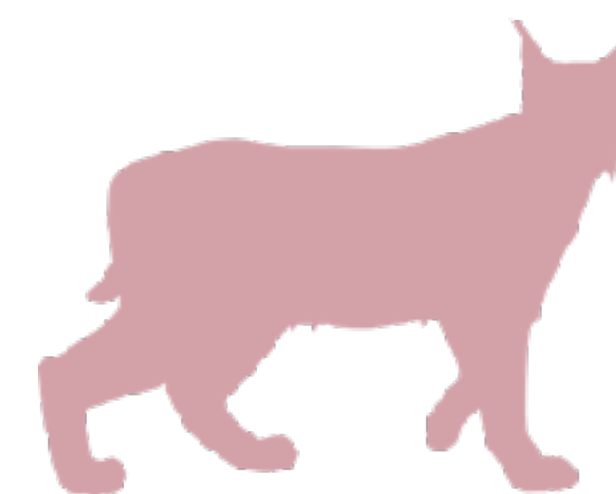
- Autoregressive models may be fine for forecasting in some cases.
 - They are not suited for drawing causal interpretations.
-
- Instead, we'll model the population dynamics using ordinary differential equations.

Lotka-Volterra model

$$\frac{dH}{dt} = H_t \times (\text{birth rate}) - H_t \times (\text{death rate})$$



$$\frac{dL}{dt} = L_t \times (\text{birth rate}) - L_t \times (\text{death rate})$$



(Lotka, 1925; Volterra, 1926)

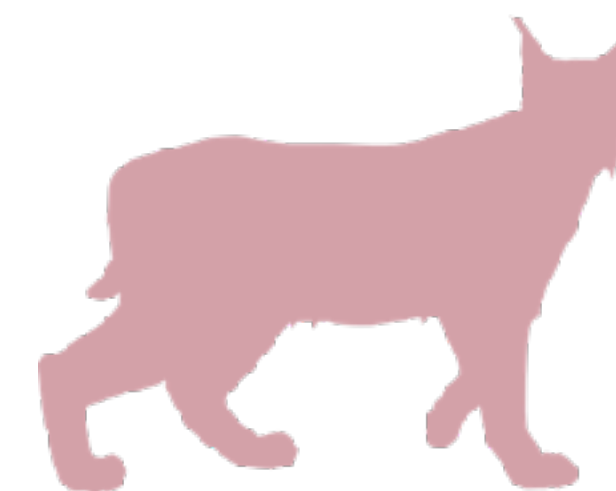
$$\frac{dH}{dt} = H_t b_H - H_t(L_t m_H)$$

*birth rate
of hares*

*impact of lynx
on hares*



$$\frac{dL}{dt} = L_t \times (\text{birth rate}) - L_t \times (\text{death rate})$$



$$\frac{dH}{dt} = H_t b_H - H_t(L_t m_H)$$

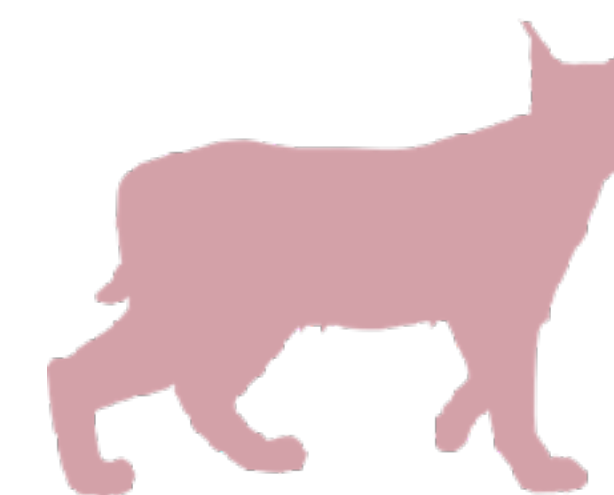
*birth rate
of hares*

*impact of lynx
on hares*



$$\frac{dL}{dt} = L_t(\underline{H_t b_L}) - L_t m_L$$

*birth rate of lynx
depends upon hares*



**oversimplified example b/c*

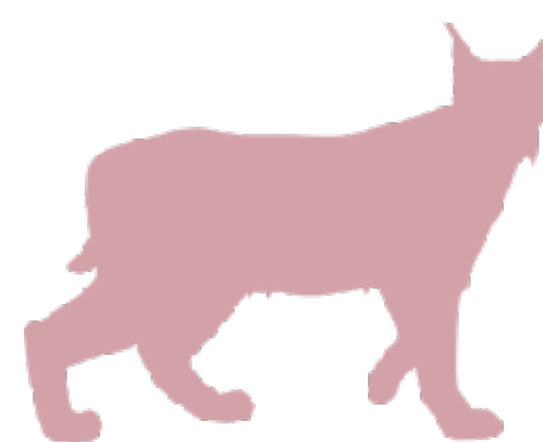
- hares die in ways other than being eaten by lynx
- lynx eat things other than hares



$$h_t \sim \text{LogNormal}(\log(p_H H_t), \sigma_H)$$

$$\frac{dH}{dt} = H_t b_H - H_t(L_t m_H)$$

$$H_T = H_1 + \int_1^T \frac{dH}{dt} dt$$



$$l_t \sim \text{LogNormal}(\log(p_L L_t), \sigma_L)$$

$$\frac{dL}{dt} = L_t(H_t b_L) - L_t m_L$$

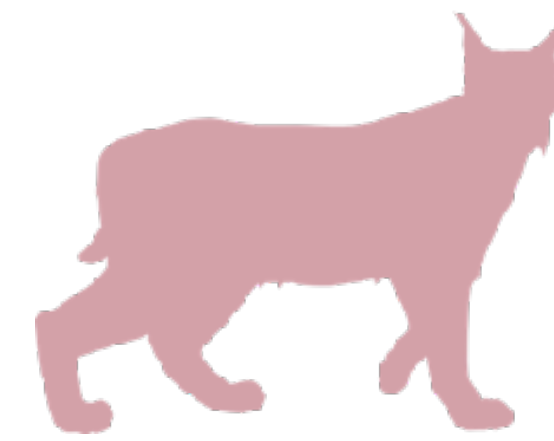
$$L_T = L_1 + \int_1^T \frac{dL}{dt} dt$$

*observed
hare pelts*



$$h_t \sim \text{LogNormal}(\log(p_H H_t), \sigma_H)$$

*observed
lynx pelts*



$$l_t \sim \text{LogNormal}(\log(p_L L_t), \sigma_L)$$

$$\frac{dH}{dt} = H_t b_H - H_t(L_t m_H)$$

$$H_T = H_1 + \int_1^T \frac{dH}{dt} dt$$

$$\frac{dL}{dt} = L_t(H_t b_L) - L_t m_L$$

$$L_T = L_1 + \int_1^T \frac{dL}{dt} dt$$

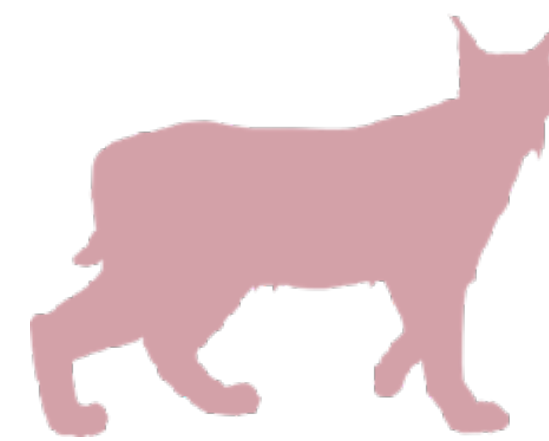
*pH and pL are the probabilities that hares and lynx are caught by trappers

*observed
hare pelts*



$$h_t \sim \text{LogNormal}(\log(p_H H_t), \sigma_H)$$

*observed
lynx pelts*



$$l_t \sim \text{LogNormal}(\log(p_L L_t), \sigma_L)$$

$$\frac{dH}{dt} = H_t b_H - H_t(L_t m_H)$$

$$H_T = H_1 + \underbrace{\int_1^T \frac{dH}{dt} dt}_{\text{cumulative changes in } H \text{ until time } T}$$

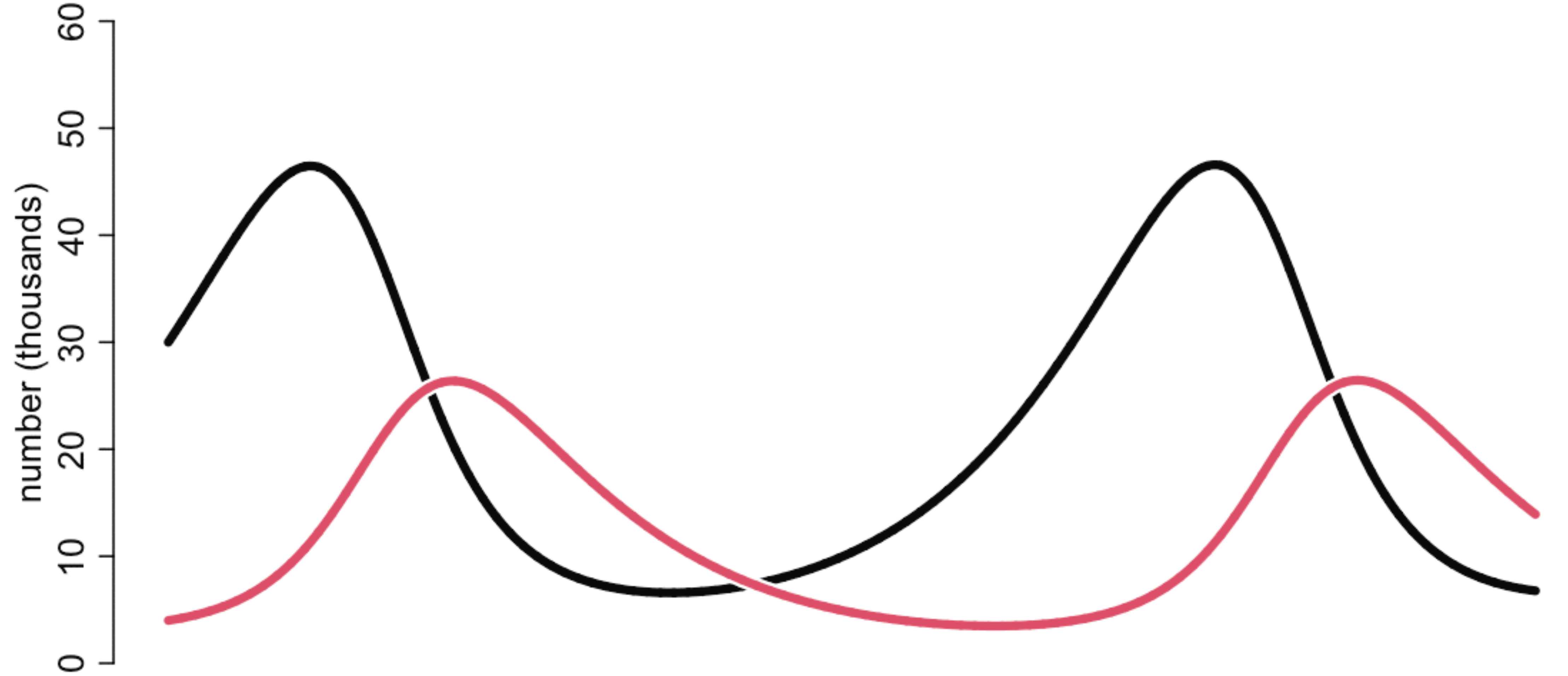
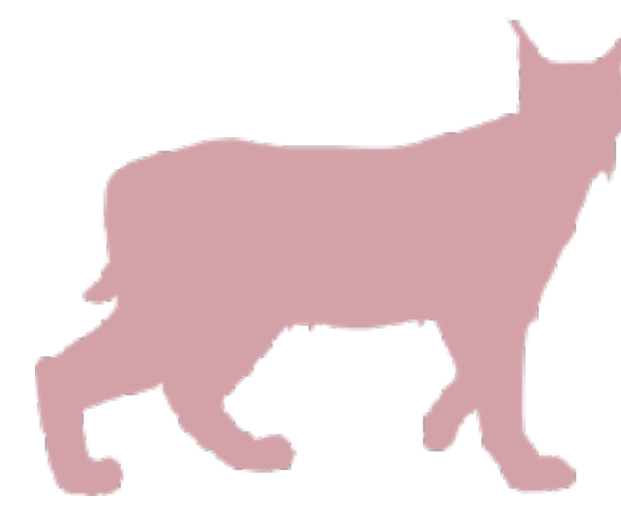
*cumulative changes
in H until time T*

$$\frac{dL}{dt} = L_t(H_t b_L) - L_t m_L$$

$$L_T = L_1 + \underbrace{\int_1^T \frac{dL}{dt} dt}_{\text{cumulative changes in } L \text{ until time } T}$$

*cumulative changes
in L until time T*

Prior Simulation




```

functions {
  real[] dpop_dt( real t,           // time
                  real[] pop_init,  // initial state {lynx, hares}
                  real[] theta,     // parameters
                  real[] x_r, int[] x_i) { // unused

    real L = pop_init[1];
    real H = pop_init[2];
    real bh = theta[1];
    real mh = theta[2];
    real ml = theta[3];
    real bl = theta[4];
    // differential equations
    real dH_dt = (bh - mh * L) * H;
    real dL_dt = (bl * H - ml) * L;
    return { dL_dt , dH_dt };
  }
}

data {
  int<lower=0> N;           // number of measurement times
  real<lower=0> pelts[N,2]; // measured populations
}

transformed data{
  real times_measured[N-1]; // N-1 because first time is initial state
  for ( i in 2:N ) times_measured[i-1] = i;
}

parameters {

```

```

functions {
  real[] dpop_dt( real t,           // time
                  real[] pop_init,  // initial state {lynx, hares}
                  real[] theta,     // parameters
                  real[] x_r, int[] x_i) { // unused

    real L = pop_init[1];
    real H = pop_init[2];
    real bh = theta[1];
    real mh = theta[2];
    real ml = theta[3];
    real bl = theta[4];
    // differential equations
    real dH_dt = (bh - mh * L) * H;
    real dL_dt = (bl * H - ml) * L;
    return { dL_dt , dH_dt };
  }
}

```

*Computes
cumulative
change to time t*

```

data {
  int<lower=0> N;           // number of measurement times
  real<lower=0> pelts[N,2]; // measured populations
}
transformed data{
  real times_measured[N-1]; // N-1 because first time is initial state
  for ( i in 2:N ) times_measured[i-1] = i;
}
parameters {

```

```

functions {
  real[] dpop_dt( real t,           // time
                  real[] pop_init,  // initial state {lynx, hares}
                  real[] theta,     // parameters
                  real[] x_r, int[] x_i) { // unused

    real L = pop_init[1];
    real H = pop_init[2];
    real bh = theta[1];
    real mh = theta[2];
    real ml = theta[3];
    real bl = theta[4];

    // differential equations
    real dH_dt = (bh - mh * L) * H;
    real dL_dt = (bl * H - ml) * L;
    return { dL_dt, dH_dt };
  }
}

data {
  int<lower=0> N;           // number of measurement times
  real<lower=0> pelts[N,2]; // measured populations
}

transformed data{
  real times_measured[N-1]; // N-1 because first time is initial state
  for ( i in 2:N ) times_measured[i-1] = i;
}

parameters {

```

***Computes
cumulative
change to time t***

$$\frac{dH}{dt} = H_t b_H - H_t(L_t m_H)$$

$$\frac{dL}{dt} = L_t(H_t b_L) - L_t m_L$$

***<- factored out H and L to reduce
multiplication steps***


```

}
parameters {
  real<lower=0> theta[4];      // { bh, mh, ml, bl }
  real<lower=0> pop_init[2];   // initial population state
  real<lower=0> sigma[2];      // measurement errors
  real<lower=0,upper=1> p[2];  // trap rate
}

```

```

transformed parameters {
  real pop[N, 2];
  pop[1,1] = pop_init[1];
  pop[1,2] = pop_init[2];
  pop[2:N,1:2] = integrate_ode_rk45(
    dpop_dt, pop_init, 0, times_measured, theta,
    rep_array(0.0, 0), rep_array(0, 0),
    1e-5, 1e-3, 5e2);
}

```

```

model {
  // priors
  theta[{1,3}] ~ normal( 1 , 0.5 );    // bh,ml
  theta[{2,4}] ~ normal( 0.05, 0.05 ); // mh,bl
  sigma ~ exponential( 1 );
  pop_init ~ lognormal( log(10) , 1 );
  p ~ beta(40,200);
  // observation model
  // connect latent population state to observed pelts
  for ( t in 1:N )
    for ( k in 1:2 )
      pelts[t, k] ~ lognormal( log(pop[t, k]*p[k]), sigma[k] );
}

```

***Compute
population state
for each time***

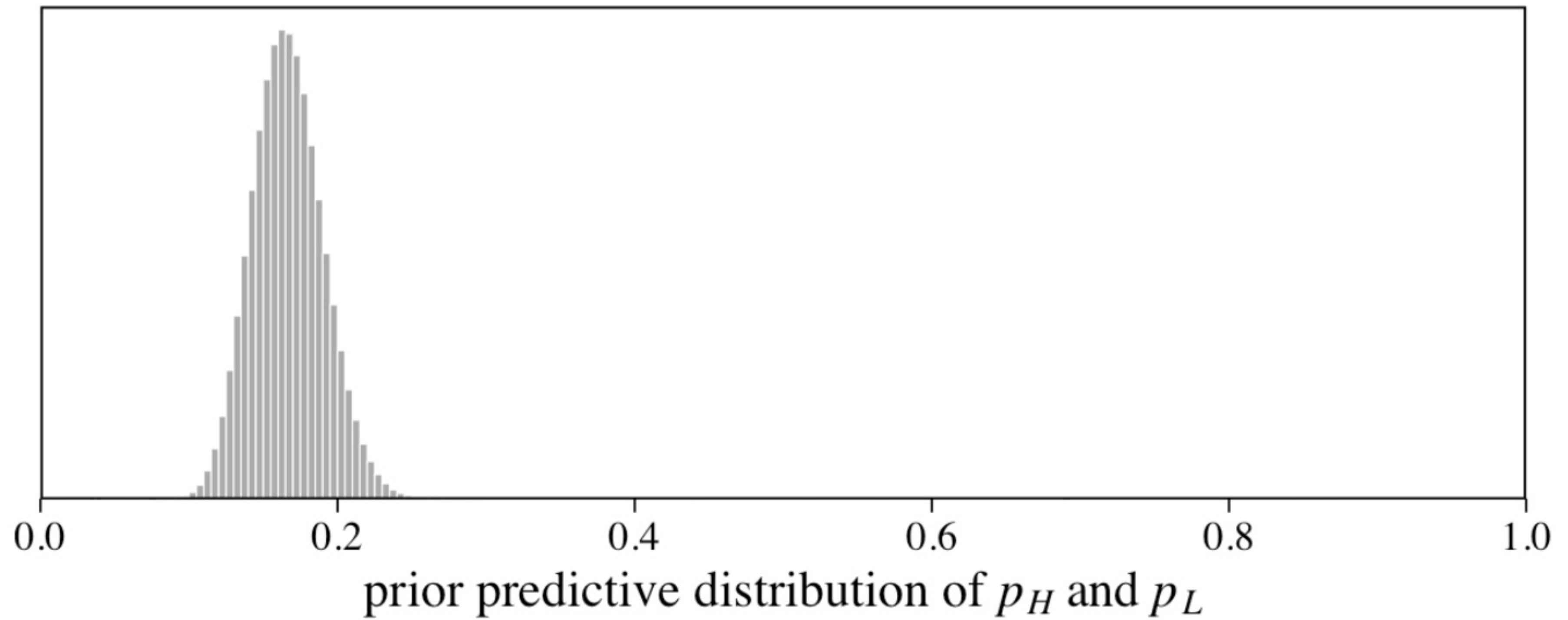
```

pop[1,1] = pop_init[1];
pop[1,2] = pop_init[2];
pop[2:N,1:2] = integrate_ode_rk45(
    dpop_dt, pop_init, 0, times_measured, theta,
    rep_array(0.0, 0), rep_array(0, 0),
    1e-5, 1e-3, 5e2);
}
model {
    // priors
    theta[{1,3}] ~ normal( 1 , 0.5 );    // bh,m1
    theta[{2,4}] ~ normal( 0.05, 0.05 ); // mh,b1
    sigma ~ exponential( 1 );
    pop_init ~ lognormal( log(10) , 1 );
    p ~ beta(40,200):
    // observation model
    // connect latent population state to observed pelts
    for ( t in 1:N )
        for ( k in 1:2 )
            pelts[t,k] ~ lognormal( log(pop[t,k]*p[k]) , sigma[k] );
}
generated quantities {
    real pelts_pred[N,2];
    for ( t in 1:N )
        for ( k in 1:2 )
            pelts_pred[t,k] = lognormal_rng( log(pop[t,k]*p[k]) , sigma[k] );
}

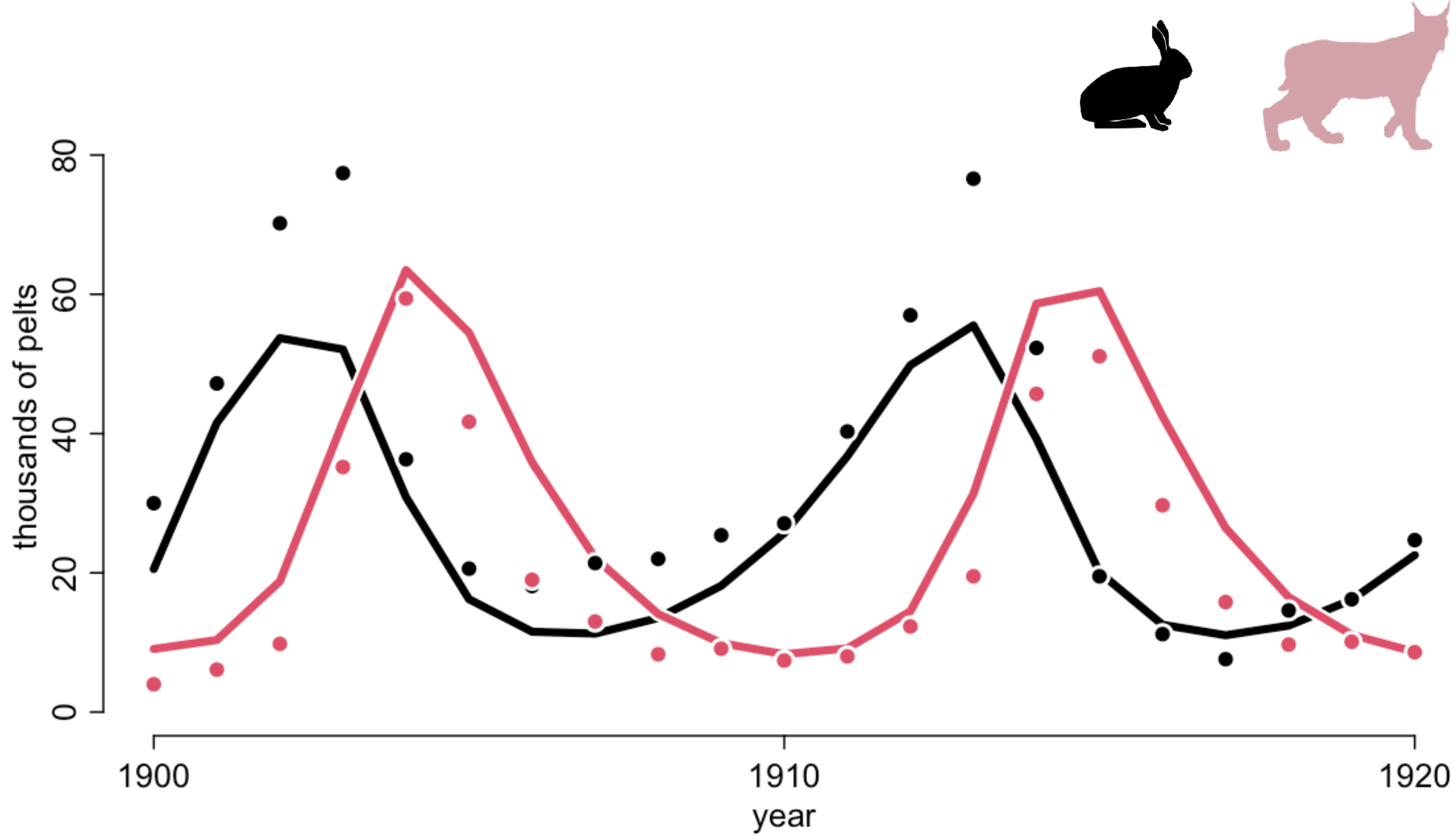
```

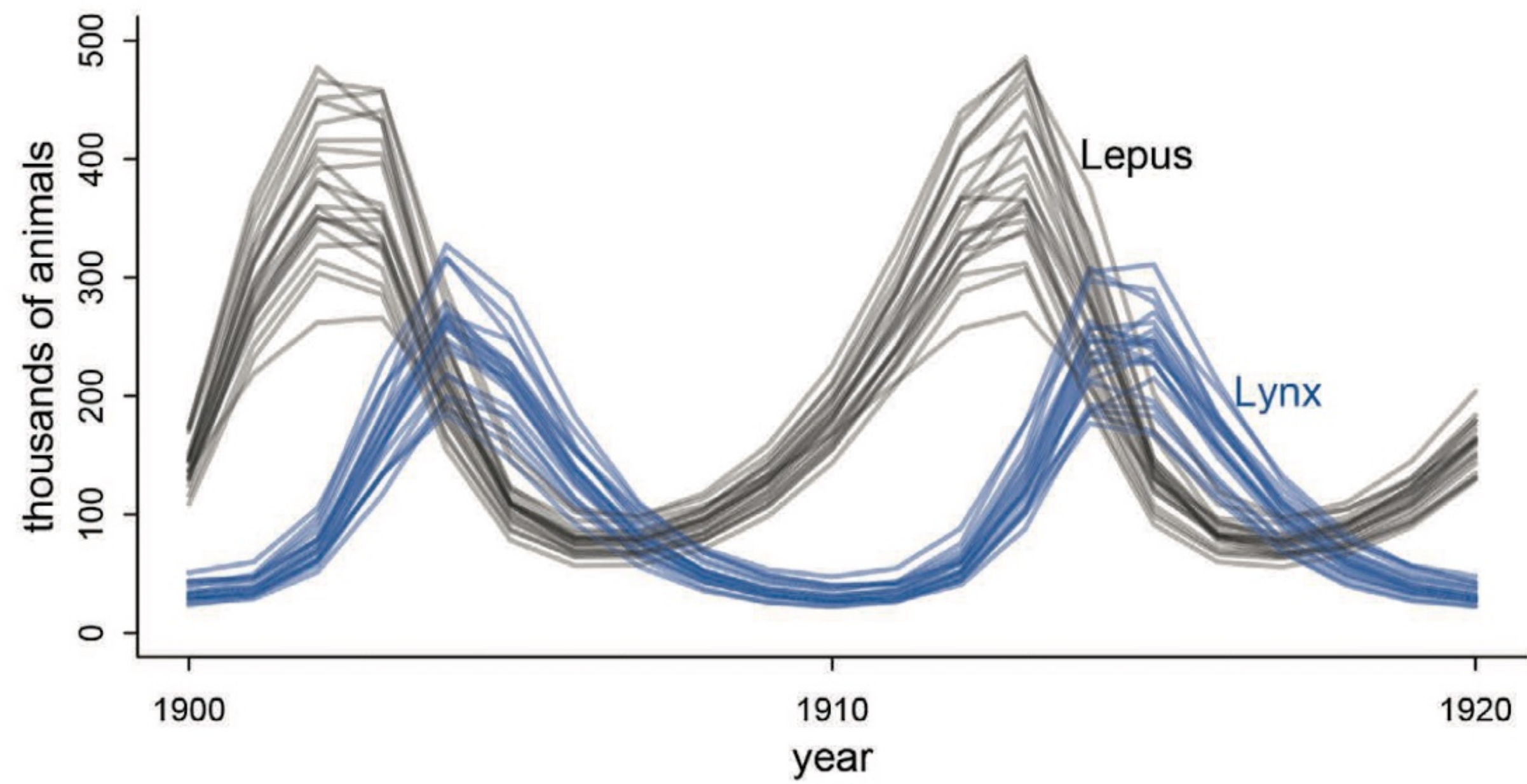
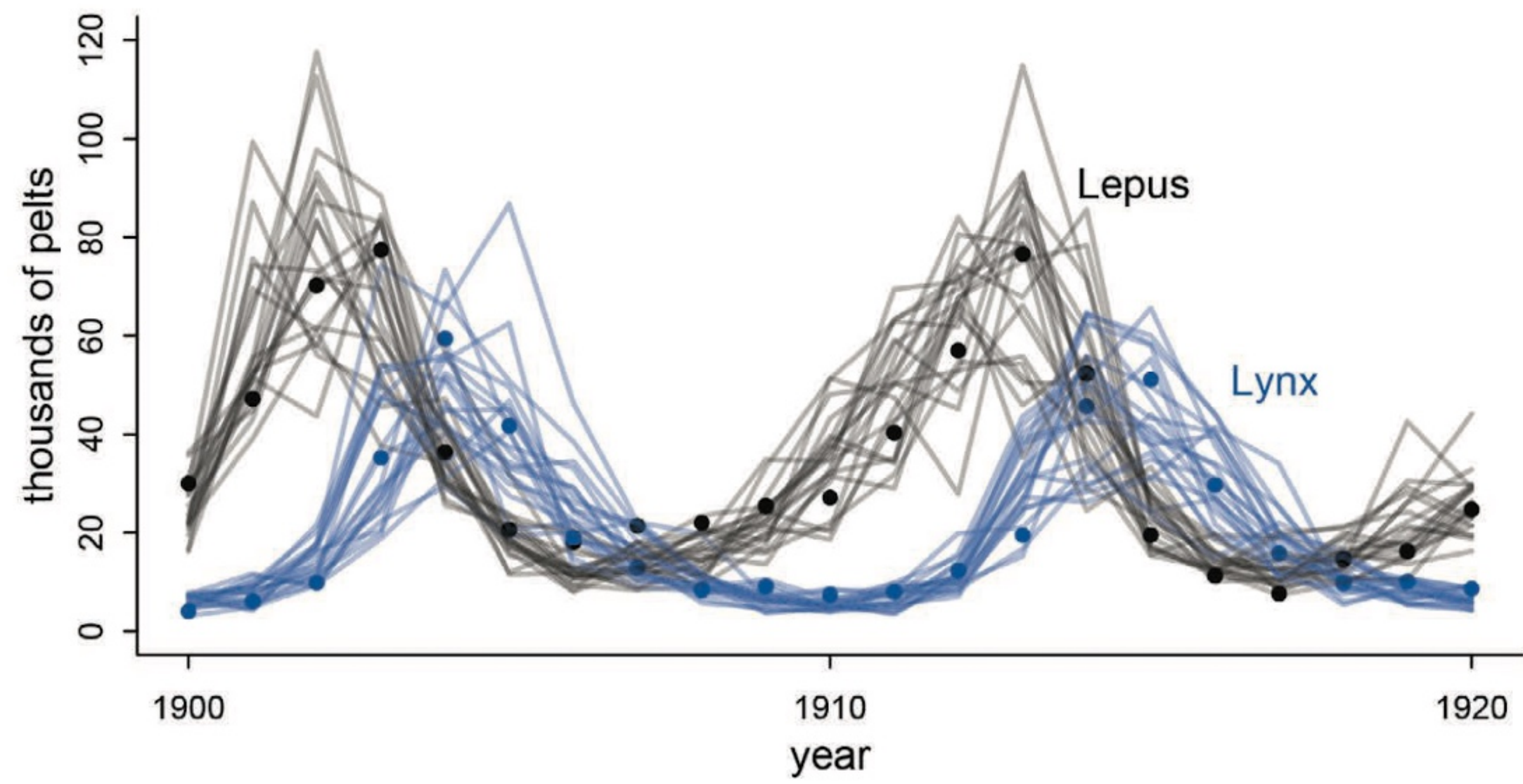
*Probability of
data, given
latent population*

1,000,000 draws from Beta(40, 200)



The Beta(40, 200) prior suggests an average trapping rate near 16%.



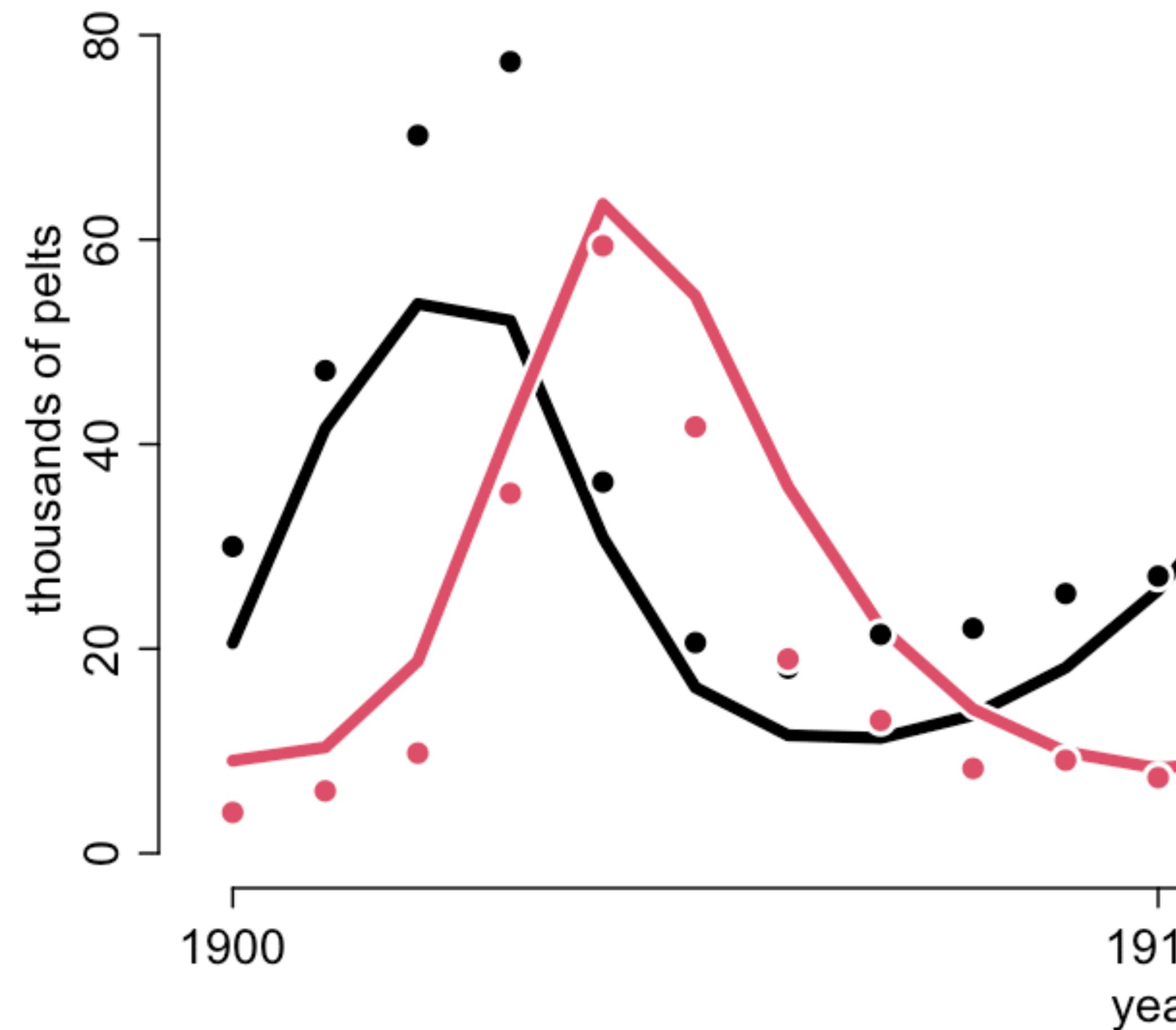


Population Dynamics

Ecologies much more complex

Other animals prey on hare

Without causal model, little hope to understand interventions



Science Before Statistics

Epicycles get you only so far

Scientific models also flawed, but
flaws are more productive

Theory necessary for empiricism

Be patient; mastery takes time;
experts learn safe habits



Student learning differential equations